

Auditable adversarial review for AI-in-the-loop decisions: a pre-registered null redirects an AI-deliberation study from better answers to governance by verifiable challenge — an empirical report on SPARRING’s Resolver

SPARRING LLM-judge pilot — V2 empirical report Status: Preprint, 2026-07-08 — not peer-reviewed. The V2 rater numbers are final (frozen dataset, 2026-06-26). The governance evidence folded in below is **directional, first-signal:** the mechanism audit and the §6 catch-rate are LLM-coded; the seeded-defect *grounding* is machine-checked (Path-A, §8), with legitimacy expert-pending. Read it with the validation stance (§11.1) and the limitations (§11.2). The body’s citations were machine-verified — resolved against external indexes and judged for correct application by a two-vendor pipeline — with **no hallucinations and no wrong-paper** (13 existence independently confirmed, 4 application evidence-grounded). **Pilot:** `sparring-llm-judge-pilot-2026-05-03-v2`. Pre-registration: locked at tag `v2-prereg-2026-05-03` (2026-05-03); public archive and DOI in §13. Analysis code + frozen results: [scripts/analyze-v2-primary.php](#), [analysis/primary-results.json](#). Governance evidence: mechanism audit [docs/spars/mechanism-audit-2026-06-26/](#); seeded-defect grounding pilot [pilots/seeded-defect-2026-06-26/](#); the grounding verifier [pilots/seeded-defect-2026-06-26/scripts/](#).

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Abstract

In plain terms (reader’s summary). We started by running a blind taste test. For two real decisions we showed people two answers — one from a single AI pass, one from the full SPARRING process — without saying which was which, and asked which was better. People did **not** prefer the SPARRING answer; mostly they called it a tie. Two cautions ride with that: the raters barely agreed *with each other*, so the test was too noisy to detect a real difference either way (“no difference found” means “this test couldn’t tell,” **not** “SPARRING is no better”); and, the real point, a blind preference over a *finished* answer is the wrong tool for SPARRING to begin with — its value lives in the **process**, the way a kitchen’s worth is in catching the undercooked dish, not in how the finished plate looks to a stranger. So the null did not end the inquiry; it **redirected** it. Following the redirect, we asked a different and harder question: not “*are the answers better?*” but “*is the reasoning good governance — can you audit it, is it actually grounded rather than confident-sounding theater, and does it catch things that matter?*” On that question the signal is positive. The second agent caught real, checkable defects that would otherwise have shipped (an authorization stub that **failed open**; an account-lockout rule that was a denial-of-service trap). Its critiques came with checkable receipts about **96%** of the time — far more than a plain single pass, and modestly more than even a genuinely tough skeptic who pushes back hard and grounds most of its own concerns too. And we built a tool that **independently checks those receipts**: it confirms a cited source actually

exists — by machine, no AI involved — and separately flags when a real source is being used to prop up a claim it doesn't support. The key part needs no AI's say-so and no subject expert: anyone can confirm the source is real. The contribution, then, is not that SPARRING writes an answer that *looks* better — shown the two finished answers cold, people didn't prefer the SPARRING one. It is that the answer arrives **governed**: reasoning that is *auditable, grounded, and consequential* — one you have grounds to *trust and act on*. The grounding is real, and the parts you can't check just by reading the documents — security flaws, citations — a machine can check for you. Not a better answer on sight; a better one to act on.

SPARRING is a deliberation method that grew out of a concrete operational failure: an agreeable AI in the decision loop, whose bias *compounds* across multi-agent workflows as one agent's sycophancy becomes another's input. It was built — by abstraction from the Socratic method and the generative adversarial network — to make confident wrongness structurally hard to ship. This report measures one of SPARRING's two components — the **Resolver**, its adversarial challenge; the **Proposer** front end (SPARK → Pattern Lock → the Cut) is a separate forthcoming report, so the claims here are scoped to the Resolver, not to SPARRING as a whole. The founding concern was trust in the *reasoning*; but to test it we reached for the question a skeptic asks first. We set out to answer a simple version of it: when an AI is given a real decision, does running it through SPARRING — a two-role process where one agent proposes and a second, independently-grounded agent is *required* to push back before either can agree — produce a recommendation that people judge to be better than the same AI answering in one pass? (An earlier, looser pilot had found raters *did* prefer the SPARRING answer; the tighter, pre-registered study reported here finds no such preference; we do not stand on the earlier result (§5.2).) Pre-registered, blind, two decision cases, ten raters, plus a cross-vendor panel of three AI judges. The answer was **no measurable difference**: raters did not lean toward the SPARRING answer (if anything slightly the other way, mostly tied), no criterion reached significance, and the AI-judge panel did not track human judgment well enough to pass our pre-set agreement bar. The raters also barely agreed *with each other* (agreement near chance), which caps how much any of these numbers can carry.

That null is **load-bearing, and it stays in the paper** — but as the honest turn that earned a reframe, not as the finding. Read correctly, it says output-preference is the wrong instrument for a *process* method: five of our six rubric criteria were about the *quality of the reasoning*, yet to hold the blind we flattened away the very process evidence those criteria needed (§5). Following that redirect, this report assembles the evidence the blinded study could not see, organized around three properties of good governance. **Consequential**: an audit of real SPARRING ceremonies finds the second agent was load-bearing — it changed the decision or caught a defect that would have shipped — in an adversarially **re-judged** two-thirds of the recorded cases (LLM-coded; §6.2), with five documented real-world catches (3 engineering, 1 product, 1 marketing; §6.1); and in a controlled test a challenger recovered most of a producer's *own* would-ship flaws — for some producing models, materially more than the producer's self-review caught (§6.3). Iterating that challenge *to agreement* — the both-must-agree gate — was itself put to a pre-registered controlled test and produced no cleaner output than a single challenge: a **second null**, on the gate's marginal quality, consistent with this paper's central turn (§6.4). **Grounded, not theater**: a seeded-defect pilot finds SPARRING backs **96%** of its concerns with checkable artifacts — far above a raw single pass (58%) and modestly above even a *realistic, competent skeptic* (83%), at **no precision cost** (§7). The edge over a strong skeptic is small and concentrates where checking requires reaching outside the documents; but the load-bearing point is not the size of that gap — it is that the grounding is **machine-checked for existence and cross-vendor-judged for application, not a single model grading itself**. Me-

chanically checkable: a two-layer verifier confirms each cited artifact *exists* against public registries (MITRE, Crossref, the IETF datatracker — security and scholarship here, but the mechanism is domain-general, §8) with no model in the loop, and a dual-vendor judge flags backward-cited sources — now across all four defect families, not only the external slice, and (exploratory) extended from the Challenger’s citations to the answer’s own load-bearing figures (§8) — making grounding checkable without a domain expert. The thesis the data supports is therefore not an answer a blind reader prefers, but a **governed** one — auditable, grounded, consequential reasoning you can trust and act on (§9). The redirect was driven by the **output-preference null** (§4); the later finding that iterating to agreement buys no cleaner answer (§6.4) neither produced the governance value nor undercuts it — that value rests on the single grounded challenge, and would stand whether or not iterating had helped.

Honest-claim tag (SPARRING house rule): every load-bearing claim below grounded only in SPARRING’s own documents is marked *internally consistent / externally unvalidated*. The model-coded evidence — the §6 catch and base-rate results, and the *legitimacy* coding behind §7’s grounding rates — is **directional, first-signal**: strong enough to redirect the inquiry and motivate confirmatory work, not strong enough to settle it. The *verifiable* parts carry the weight: the real CWEs, the verifier’s machine-checked grounding (now across all four defect families), and the real would-have-shipped catches a reader can confirm directly; the model-coded rates ride as supporting-and-caveated (§9, §11).

Keywords: LLM-as-judge · self-correction · multi-agent · AI governance · sycophancy

1. Introduction

1.1 The problem: trusting an AI you have put in the decision loop

A growing share of real decisions now passes through an AI at some point — drafting the options, weighing the evidence, writing the recommendation. The open problem underneath that shift is **evaluation**: how do you tell whether an AI-assisted decision process is actually better, rather than just more confident or more elaborate? It is one of the hardest open problems in the agentic space, and it is the one this study walked into.

1.2 Where SPARRING came from

SPARRING was not designed in the abstract as a research artifact; it precipitated out of real work running agentic software, where an AI was already in the decision loop. The failure showed up first as a *single-agent* problem — an agreeable model gives confident, accommodating editorial feedback — but it turned **structural** once the author moved to **multi-agent** workflows: when one agent’s sycophantic output becomes another agent’s input, the bias *appeared to compound* rather than merely persist — in the author’s experience, each pass tended to ratify and amplify the last. That pattern recurred across more than one agentic ecosystem, which is what made it look like a property of the *architecture* rather than a quirk of one model, and what made a structural fix worth reaching for.¹

¹The concrete provenance behind the abstractions in this paragraph, for the record: the single-agent failure first surfaced in the author’s in-house editorial-comment workflow for **BioDigital Novels**; the multi-agent **amplification** appeared after introducing multiple-agent workflows in **LineMind Novelist**; the same pattern recurred in **Lifspel**, where

The author’s response was a single discipline, conceived as an abstraction of two older ideas — the **Socratic method** (interrogation that exposes unfounded assumptions) (Vlastos 1983) and the *shape* of a **generative adversarial network** (GAN — a generator improved by an adversary that scores its output, minus the training machinery) (Goodfellow et al. 2014) — distilled into a rule an agent could actually follow: *be polite, not pleasing* (PNP). Require the second agent to genuinely push back rather than smooth the road. That parentage is why the two roles are adversarial *by construction*, not a politeness checklist bolted on. PNP was wired into a production workflow about a month before anything called SPARRING existed; the author found it markedly effective in use, and his two collaborators adopted it almost immediately — an anecdotal signal, of exactly the kind that motivates the study below rather than substituting for it. Using it surfaced the need for a structured opening move (the **SPARK** phase), and then the fuller four-phase loop; only at that point did the method get its name. The loop was built **backward from PNP** — discovered by operating it, not theorized and then deployed — and hardened through real use across several codebases (the same body of ceremonies §6 audits), which is where its specific commitments earned their place: deliberately divergent evidence bases for the two roles, a both-must-agree gate the proposer cannot close alone, and a verifiable-artifact rule that makes a concern count only if it cites something checkable. That the challenger came first — the rest of the ceremony grew around it, not the reverse — is design history, not evidence; but it is why the controlled study later isolates that single construct (§6.3).

The reflex PNP names is usually called sycophancy (Sharma et al. 2023), but it is only the most-cited member of a *family* of compounding failure modes an AI-in-the-loop exhibits — confirmation bias, anchoring, misread questions, specialization blind spots, hallucinated detail, confidently-wrong outputs, bandwagon contamination. SPARRING’s target is the whole family: not “make the agent nicer to argue with” but “make it structurally hard for unchallenged, confident wrongness to reach a decision.” Note what kind of problem that is. At root it is not “*are the answers better?*” It is “**can I trust the reasoning of an AI I have put in my decision loop?**” — a *governance* question, present at the method’s birth.

1.3 Why it earned a study — and the instrument we reached for first

By early 2026 SPARRING was a working method with a documented structure and a deliberately honest meta-claim attached: its value was *structurally argued but not empirically measured* — a **testable hypothesis, not a proven result** (a discipline the method imposes on its own claims; see the honest-claim tag throughout). It had worked anecdotally and spread by word of use (§1.2) — which is precisely the kind of signal that *demand*s a rigorous test rather than standing in for one. That gap is what earned it a study. The hard part was choosing the instrument. The founding concern was trust in the *reasoning*, but the obvious, legible, externally-credible test is the one a skeptic asks for first: never mind the process — *are the answers any better?* That is the test we reached for, and ran carefully.

1.4 What this report finds

We walked in with that framing — *better answers* — and the data walked us back out of it. The redirect is the spine of this paper: from “does this produce better answers?” back to the question SPARRING was built to address — “**is the reasoning good governance?**” By governance we mean

the author implemented the first basic version of PNP. All three are the author’s own products. PNP was adopted by the author’s two collaborators on these projects shortly after.

three concrete, separable properties: the reasoning is **auditable** (you can *check whether* what the second agent did is grounded, not merely see that it happened), **grounded** (its critique stands on checkable evidence rather than confident assertion), and **consequential** (it changes decisions or catches defects that would otherwise ship). This is not a marketing reframe applied after a disappointing result — though a reader is right to demand more than our say-so on that, and §5 supplies it from the frozen data. The redirect is also, secondarily, a *homecoming*: governance was the founding concern (§1.2), output-preference the detour. But the case for following it rests on the §5.1 evidence, not on that origin story — and the rest of the paper is the evidence that the founding concern, measured with the right instruments, holds up.

§3–§4 report the output-preference study and its null. §5 reads the null honestly and shows why a blinded output-comparison is structurally unable to reveal what SPARRING does. §6–§8 are the redirect made concrete: the process is consequential (§6), grounded rather than theater (§7), and its grounding **machine-checked** — existence checked against external registries with no model, correct-application judged cross-vendor — across all four defect families, not just the externally-cited slice (§8). §9 states what this establishes — governance — and what it pointedly does not. (Two version conventions run through the paper and do not share a ladder: capital **V1/V2** are the two runs of the output-preference rater study above; the pilots in §6–§8 carry their own lowercase iteration numbers — the seeded-defect pilot’s v2/v3, the catch-rate pilot’s v3 — internal to each pilot, not further steps in the V-sequence.)

2. SPARRING in brief

Strip the names off and the method is this: one role drafts a recommendation; a second role, working from a deliberately different body of evidence, has to attack it — and an attack only counts if it points at something checkable (a specific source, file, study, or number), not a vibe. As first built — and as the name pictures it — they go back and forth, and the process is only allowed to finish when **both** roles say “I agree” — the proposer can’t declare victory alone. If they run out of rounds without agreeing, that disagreement is reported, not smoothed over. (Hold that loop lightly: a pre-registered test (§6.4) found its extra rounds buy no cleaner output than one challenge, so the form this paper lands on — laid out in §2.1 — runs the challenge once and stops.)

In SPARRING’s own vocabulary, those two roles are the **Generator** and the **Challenger**, and the “must cite a checkable thing” rule is the *verifiable-artifact* discipline. The point of the gate — both-must-agree, non-bypassable — is the part engineered for AI: a pleasing agent will *fake* agreement to be helpful, so the method makes faked agreement structurally hard rather than trusting the agent not to. That same verifiable-artifact discipline is what later makes the reasoning *auditable* and the grounding’s *existence* mechanically checkable (§7–§8) — the governance properties are designed into the method, not read into it afterward. (The worked examples run on software/security, scholarship, and standards because those carry public registries to check against; the check itself keys off each citation’s *type* and its majority resolves against the decision’s own briefing, so the mechanism is domain-general — §8.) The two roles diverge by *evidence base* by default; for high-consequence decisions the divergence can extend to the **model substrate** — Generator and Challenger run on different vendors — so the two are less likely to share trained-in pleasing patterns (an author-reported practice for high-consequence decisions, though this paper’s pilots ran single-substrate; §11.2).

SPARRING did not invent structured disagreement. It sits in an old family — devil’s advocacy, dialectical inquiry, the premortem, analysis-of-competing-hypotheses (Schwenk 1990; Klein 2007;

Heuer 1999) — and its own design lineage is narrower and more mechanical: the Socratic method’s keep-asking-until-it-breaks interrogation and the generator-versus-critic shape of a GAN (§1.2), which is why its two roles are adversarial-by-construction rather than a checklist. *Internally consistent / externally unvalidated*: its claimed distinctive niche is a **verifiable, non-bypassable adversarial challenge** built for the AI-in-the-loop case — a concern counts only if it cites something checkable, and the challenge runs behind a gate the proposer cannot close alone. This report’s subject is that **Resolver**, and the evidence lands its value on the **single grounded pass** (§6.3–§8): one challenge catches the producer’s own would-ship blind spots, and its critique is mechanically checkable. The both-must-agree gate that would iterate that challenge *toward* a revised, agreed output — or report the disagreement — is the enforcement structure, not the source of the value: its controlled test (§6.4), isolated on a net-flaws-remaining measure, finds that iterating to agreement adds no output-quality gain over a single challenge — a pre-committed null. The name is a loan worth paying plainly: *sparring* pictures repeated exchange, but the evidence here puts the value in the **quality of a single grounded challenge, not the count of rounds** — the back-and-forth is the safe frame the method runs in, not where the leverage lives. SPARRING’s **Proposer** — the SPARK → Pattern Lock → the Cut front end that produces the proposal the Resolver works on — is out of scope here: the subject of a separate forthcoming report.

2.1 The Resolver, initial process → final process

This report is partly the story of a product that changed shape when its own test told it to — specifically, its central move, retiring an *iterate-to-agreement* loop, was made because a pre-registered test found no gain from the extra rounds (§6.4); the parts that replaced the loop are design choices still under confirmatory test (§12). Two versions of the Resolver bracket the work — the loop as first built, and the loop the evidence arrived at — and one thread of what follows is the bridge between them. Both are laid out here so every later result and limitation can point back to the exact stage it bears on. (The stage ids **S1–S5** below are new; they name the *final* loop’s five shipped parts. The *initial* loop’s four parts carry their own ids, **P1–P4** — proposal in, challenge/revise, the both-must-agree gate, and the answer it produced — used in Figure 1 and, where a limitation attaches to a retired part specifically, in Appendix B, for the same reason S1–S5 are: so both the figure and the attachment tags read as a map, not just an illustration. Neither ladder shares numbers with the paper’s **V1/V2** rater-study runs, the pilots’ lowercase **v2/v3** iterations, or the §6.4 study’s condition labels **R0/R1/R2/R1′** — those are separate and unrelated.)

The final form — the accountability pass: a single grounded challenge delivered with a definitiveness gate and a trust rider (§10). The figure carries the transformation; the five parts themselves are:

- **S1 • One grounded pass** — a single cross-vendor challenge, run once; a concern counts only if it cites something checkable. Audited by the grounding verifier (§8: machine existence check + cross-vendor application).
- **S2 • Definitiveness gate** — the correctness check: every hard specific must trace to the given facts or be marked an estimate (PASS / REVIEW / FAIL).
- **S3 • The answer** — output part one: the clean recommendation, reasoning visible, no caveats woven through.
- **S4 • Trust rider** — output part two: what is grounded, what is assumed, what was contested; unresolved disagreement is *reported* here, not smoothed into a false agreement.
- **S5 • Anchored correction** — a flagged claim is fixed in place, not re-decided from scratch.

The test that reshaped the Resolver: iterate-to-agreement out, one grounded pass in

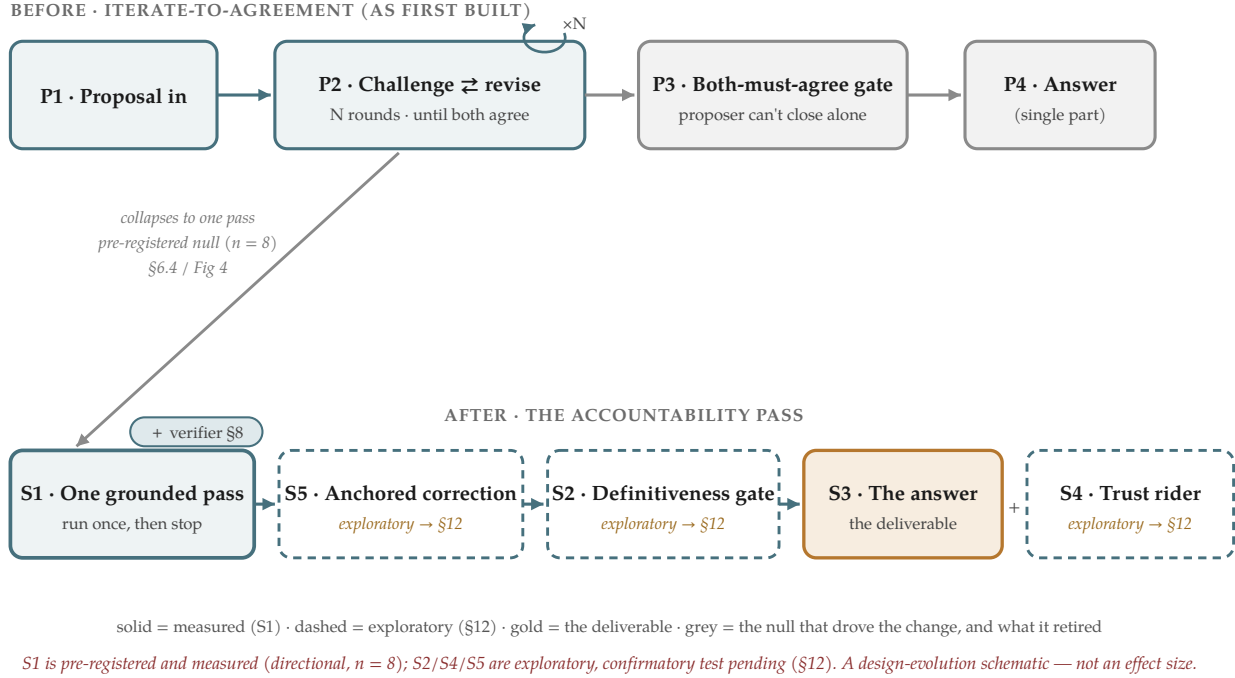


Figure 1: The test that reshaped the Resolver. The Resolver was first built as an *iterate-to-agreement* loop (top): a challenger and producer argued across rounds behind a both-must-agree gate before producing an answer. A pre-registered controlled test (§6.4; $n = 8$, directional) found the extra rounds bought no cleaner output *than a single challenge* — a null — so the loop **collapsed** to a single grounded pass that then stops (S1); the gate and the answer it produced (grey) are shown only to illustrate the shape of the original process, retired along with it. The parts that make the result checkable — the definitiveness gate (S2), the trust rider (S4), anchored correction (S5), and the grounding verifier behind S1 — were **added** (bottom). S1 is pre-registered and measured; S2/S4/S5 are exploratory (confirmatory test, §12); S3 is the answer itself. A design-evolution schematic, not an effect size.

The S-numbers label the parts; in operation they run $S1 \rightarrow S5 \rightarrow S2 \rightarrow S3 + S4$ — challenge once (S1), fix each flag in place (S5), let the gate confirm nothing is overstated (S2), then hand back the answer (S3) with its trust rider (S4).

What the evidence changed — and which node changed it.

- **Iterate-to-agreement was retired.** The pre-registered null in **J5 • Resolver-iteration** (§6.4) found iterating to agreement buys no cleaner output than a single challenge, and can churn in new flaws. The final form runs **one grounded pass and stops** (S1); the leverage was the single grounded challenge, not the rounds.
- **The both-must-agree gate left the operating form.** It drove the old iterate loop, so it went with it. Its one surviving job — *report disagreement rather than fake it* — moved into **S4 • the trust rider**.
- **S2–S5 are the parts that replaced it:** the definitiveness gate, the two-part answer + trust rider, and anchored correction (all **J5**, §8.1 / §10), plus the grounding verifier that makes S1 auditable (**J7**, §8). Together they convert “an agreed answer” into “an answer you can audit and know how far to trust.”

The honest split: **S1 is established** (pre-registered, measured, though directional — §11); **S2, S4, S5 are exploratory** (not pre-registered; confirmatory test named in §12); **S3** is the deliverable.

2.2 The journey map — how the findings were established

The findings did not arrive in the product’s order; they arrived in an *exploration* order, and knowing which is which lets a reader see what rests on what. Seven nodes make up that path. Later sections tag each claim and limitation with the node it surfaced in (**J#**) and the Resolver stage it bears on (**S#**), citing both the id and its short title so the attachment is legible without a lookup — e.g. each §11.2 limitation ends with a fixed tag: *Attaches to: S# • title — Surfaced in: J# • title — one plain sentence of what it means.*

J1 → J2 → J3 → J4 → J5 → J6 → J7

↑

J5 is the DESIGN hinge - the pivot of the Resolver's shape, distinct from the paper's argumentative hinge (the §4 output-preference null, §9). Its pre-registered null (§6.4) turned the initial iterate-to-agreement loop into the final accountability pass.

Node	Where	What it surfaced	Bears on
J1 • Output-preference study	§3–§4	the pre-registered null — a blind taste-test cannot see a <i>process's</i> value	the pivot (whole Resolver)
J2 • The pivot	§5	<i>why</i> a blinded output-comparison is structurally unable to reveal what SPARRING does	reframes the whole report

Node	Where	What it surfaced	Bears on
J3 · Mechanism audit	§6.1–§6.2	the process is consequential — real would-have-shipped catches (survivorship-caveated base rate)	S1
J4 · Controlled catch-rate	§6.3	a single challenge recovers most of the producer’s own blind spots	S1
J5 · Resolver-iteration	§6.4, §8.1, §10	iterate-to-agreement is a quality null (drives initial→final); + definitiveness gate, anchored correction, two-part output (exploratory)	S1, S2, S4, S5 · retires the gate
J6 · Seeded-defect grounding pilot	§7	the critique is grounded, not theater — ~96%, at no precision cost	S1
J7 · The grounding verifier	§8, App. C	existence is machine-checkable with no model; application judged cross-vendor	S1, S2

Every “surfaced” row is directional / small-n / LLM-coded (or survivorship-bounded); the cited section carries the governing caveats. **J#** (journey nodes) and **S#** (Resolver stages) are new to this section and do not share a ladder with the paper’s **V1/V2** (rater-study runs), the pilots’ lowercase **v2/v3** (per-pilot iterations), or the §6.4 study’s conditions **R0/R1/R2/R1’**.

2.3 Notation key

Six numbering systems run through this paper, tracking six different things. None shares a ladder with any other — a claim tagged *Surfaced in: J5* and *Attaches to: S4* is saying two separate things at once (where the finding was discovered; which shipped part it bears on), not one label read two ways.

System	What it numbers	Example	Where
J1–J7	the <i>order the findings were discovered in</i> — not the paper’s section order	J5 · Resolver-iteration	§2.2
S1–S5	the <i>final</i> Resolver’s five shipped stages	S1 · one grounded pass	§2.1

System	What it numbers	Example	Where
P1–P4	the <i>initial</i> , now-retired Resolver’s four stages	P3 · the both-must-agree gate	§2.1, Fig. 1
V1/V2	the two <i>runs</i> of the output-preference rater study	V2 (this report’s data)	§3–§5
v2/v3	iterations <i>within</i> a single pilot (seeded-defect, catch-rate)	the v3 seeded-defect run	§7–§8
R0/R1/R2/R1’	the four <i>conditions</i> in the resolver-iteration study	R2 · iterate-to-agreement	§6.4

3. Methods (the output-preference study)

3.1 Pre-registration and the deviation ledger

The methodology, the analysis plan, and the decision gate were locked before any rater data was collected (git tag v2-prereg-2026-05-03, dated 2026-05-03; the original locked text is archived as pre-registration-LOCKED-2026-05-03.md (§13), with every amendment visible as its diff against the working pre-registration). We state the limitation plainly: git commit timestamps are author-settable, so the lock’s force rests on the tagged text being recorded before the rating window opened and disclosed here — not on a cryptographic timestamp; the independent replication named in §12 is the stronger check. Pre-registration guards against hindsight bias by fixing predictions before the outcomes are known — separating prediction from postdiction (Nosek et al. 2018); that protection holds only so long as deviations from the locked plan are disclosed. There were several, **all committed before any rating existed** except where noted; the full text of each lives in the pre-registration’s §12.1. The load-bearing ones:

1. **Rater pool opened (2026-05-27).** The original closed four-rater design (two method-affiliated, two not) drew zero ratings in 24 days. We opened recruitment to a convenience sample reached through the author’s LinkedIn and Facebook, self-onboarding through a public sign-up. This trades a small designed pool for scale and externality — most recruits have no stake in the result — at the cost of selection bias from the recruitment channel (the author’s own social media). Pre-rater-data.
2. **Plain-language rebuild + substrate change (2026-06-04).** The decision materials were rewritten at a non-expert reading level (every fact, number, and citation preserved and independently re-checked), which required regenerating *both* answers from the plainer materials; the generation model moved from Claude Opus 4.7 to 4.8 under **both** conditions (so the model is never the difference between them). Pre-rater-data.
3. **One criterion re-scoped (2026-06-12).** “Factuality” became “Factual accuracy” — judged against the fact-checked briefing materials rather than against the open world — removing a contradiction with the instruction to treat the briefing as given. Pre-rater-data; judge panel re-run for that criterion.
4. **Recruitment extended on a pre-committed rule (2026-06-24).** With both cases sitting at n=7–8, we set a bounded dual stop *while still blind to all ratings*: close early the moment **both** cases reach n≥10 (the top analysis tier), or hard-stop 2026-07-01 regardless — whichever

first, no further extension. Both cases reached $n \geq 10$ and the study closed **2026-06-26**. This is a deviation from the literal pre-registered close date and is disclosed as one; it keys on sample size, not on results (the reveal stayed locked), which is what keeps it clean rather than optional-stopping — peeking at the results and halting when they look favorable, a way to manufacture false positives ([Wagenmakers et al. 2012](#)).

5. **One analysis-time deviation, disclosed (this report)**. The pre-registered “primary human reference” for judge-alignment was the *unaided* subset of the method-affiliated raters. Under the realized open pool that subset collapsed to **one person** — too small to anchor a correlation. We therefore use the **unaided completer subset (n=6)** as the primary reference, report every other subset alongside, and flag this as the one deviation decided after data existed. It does not move the conclusion: the gate fails under every reference (§4.6).

3.2 Cases, conditions, and substrate

Two decision cases, both outside SPARRING’s own domain, both judge-able by a non-expert: **Case A** — which term an employee handbook should use for staff (“team members” vs. “employees” vs. “colleagues”); **Case B** — whether an intro-programming course should teach concepts-first or language-first. Each was built into a fact-checked briefing.

For each case, **Condition A** is a single Claude Opus 4.8 pass with no method applied — we’ll call it the **baseline**. **Condition B** is a full SPARRING run on Opus 4.8. Both conditions were rendered into the *same* four-section recommendation format, so a rater cannot tell them apart by shape — a blinding fix carried over from the 2026-05-03 format-symmetry amendment, which exists because the raw ceremony transcript is visually distinctive and would otherwise give the SPARRING side away on sight. (Hold onto that fix; it becomes the whole story in §5.)

3.3 The judges, the rubric, and the two instruments

Three AI judges from different vendors — Claude Sonnet 4.6, GPT-4o, Grok-3 — scored both answers on every case, each case twice with the two answers swapped left-to-right to cancel position bias (12 judge passes total). The rubric has six criteria scored 1–7: **C1** verifiable-artifact citation, **C2** substantive vs. theatrical concerns, **C3** concerns it missed, **C4** calibrated confidence, **C5a** factual accuracy against the briefing, **C5b** engagement with the source. Each rater also did a **head-to-head** pass — for each criterion, “X better / Y better / tied” — which keeps a signal even if the 1–7 scores bunch up at the top.

3.4 Raters and the decision gate

Ten people completed both cases (the “completers”). Six rated *unaided*; four used an AI assistant during rating (recorded, per pre-registration). Prior familiarity with SPARRING was recorded after submission on a five-level scale: **nine of the ten completers answered it** — six had never heard of it (F0), two knew it by name only (F1), and one had read about it (F2); **none** reached F3 or F4, so no completer had even partial style-recognition capacity. The tenth completer did not submit the post-submission familiarity form; this is a genuine missing value, disclosed and excluded — not defaulted to F0 — from the low-exposure sensitivity subset, and the gate holds with that completer excluded as well (§4.6, §11.2).

The gate is the pre-registered pass/fail rule. For each criterion we measure how closely the AI-judge consensus tracks the human reference (Spearman correlation — a value of 0.7 means the

judges and the people rank the answers in nearly the same order; 0 means unrelated; a negative value means they rank them oppositely). High-stakes criteria (C1, C2, C5a, C5b) pass at $\rho \geq 0.7$, are borderline 0.4–0.7, fail below 0.4; the two inference-heavy criteria (C3, C4) pass at $\rho \geq 0.5$, fail below 0.2. The study **passes** only if at least 5 of 6 criteria pass with none failing; **borderline** if at least 4 pass with at most one fail; otherwise it **fails**. At $n=10$ the pre-registered claim-scaling table puts us in the top “full-inferential” tier — the strongest reporting the design allows.

4. Results of the output-preference study

4.1 The gate: FAIL

Judge-consensus-vs-human agreement, per criterion (primary reference = 6 unaided completers):

Criterion	Tier	ρ (judge vs human)	Verdict
C1 verifiable-artifact citation	high	0.78	PASS
C2 substantive vs. theatrical	high	0.32	FAIL
C3 missed concerns	low	−0.50	FAIL
C4 calibrated confidence	low	0.00	FAIL
C5a factual accuracy	high	undefined ²	INDETERMINATE
C5b engagement with source	high	0.54	BORDERLINE

Aggregate: FAIL (1 pass, 3 fail, 1 indeterminate, 1 borderline). Only one criterion — whether an answer cites checkable artifacts — showed AI judges and humans clearly tracking together. The headline correlation across all 24 score-cells is **0.50**, which sounds middling but sits *below* the gate’s passing line and rests on that single strong criterion. (Note *which* criterion survived — §5 shows it is the one whose evidence the blind could not erase.)

4.2 Did anyone prefer the SPARRING answer? No.

The head-to-head pass is the cleanest read, because it survives the scores bunching at the top. Among the six unaided raters, across both cases and all six criteria, the preferences split: **SPARRING preferred 21 times, the single-pass baseline 30 times, tied 21 times** — about **29% leaning SPARRING**. The AI judges, when they had a preference at all, leaned the *other* way from the humans (toward SPARRING 9 times, baseline 4, tied 23) but mostly saw no difference. Across the full ten-rater pool the human lean was the same shape (~32% SPARRING). The pre-registered direction — that SPARRING would be preferred on most criterion-cells — **did not hold**.

The bunched 1–7 scores agree: averaged across criteria, the SPARRING-minus-baseline difference was **−0.11 points** for humans (a hair toward baseline) and **+0.08** for the judge consensus (a hair toward SPARRING) — both indistinguishable from zero on a 7-point scale.

4.3 No criterion reached significance

The pre-registered head-to-head test (two-sided exact binomial against a coin-flip, ties excluded) reached $p < 0.05$ on **0 of 6 criteria**. The pre-registered hope (H3) was ≥ 4 of 6. Even the most lopsided criterion did not clear the bar at this sample size.

²The judges scored C5a a flat 6 out of 7 on every answer — no variation, so a correlation can’t be computed. That itself is a finding: on factual accuracy the panel saw no difference anywhere.

4.4 The raters barely agreed with each other

This is the result that bounds all the others. Inter-rater agreement (Krippendorff’s α (Hayes and Krippendorff 2007), per case — on this scale $\alpha \approx 1$ means raters agree, 0 means no better than chance) was **near zero** — about -0.05 on Case A and $+0.01$ on Case B — and the same near-zero pattern shows in cross-rater rank agreement on every criterion. People were not converging on a shared sense of which answer was better; they were close to scattered. When the measuring instrument is that noisy, it cannot detect a real difference *in either direction* even if one exists. “No difference found” is partly “this instrument could not have found one,” not proof there is none to find.

4.5 The hypotheses, scored honestly

Hypothesis (pre-registered)	Result
H1 — SPARRING preferred on most cells	Not supported (humans leaned baseline, ~29% SPARRING)
H2 — gate passes or borderline	Failed (1/6 pass)
H3 — $\geq 4/6$ head-to-head significant	0/6
H4 — rater-condition asymmetry bounded	Inconclusive at this n (see note)

Note on H4. On the pre-registered rule (cross-condition Spearman $\rho \geq 0.4$ within each criterion), H4 nominally *fails* — several criteria show negative cross-condition ρ (-1.0 to $+0.94$ across the six). But with only four AI-assisted raters, each per-criterion correlation is computed on a tiny, unstable sample and cannot be read as a real condition divergence: at $n \approx 4$ the 95% confidence interval on a rank correlation spans almost the entire $[-1, +1]$ range (the Fisher-z standard error is ≈ 1.0), so the $\rho \geq 0.4$ rule is not so much *failed* as **undefined in practice** — the sample cannot localize the correlation anywhere with usable precision. We therefore report H4 as **inconclusive at this n** rather than refuted — and flag openly that this is a more lenient reading than H1–H3 receive, justified solely by the small n on the AI-assisted side, not by the result being more favorable.

4.6 It is not an artifact of how we picked the reference

The one analysis-time deviation (§3.1) — which subset of raters anchors the judge-alignment — does **not** drive the failure. The gate fails under the unaided subset ($n=6$), the full pool ($n=10$), the low-prior-familiarity subset ($n=9$ — the F0–F2 raters, excluding the one completer with no recorded familiarity value rather than defaulting them in, §11.2), the fingerprint-capable-excluded subset (a no-op here, since no completer reached F4), and the audio-excluded subset. Every reasonable way of choosing the human reference lands on FAIL.

5. The pivot: output preference is the wrong test for a process method

5.1 What we actually measured: a process rubric, blinded to the process

It is tempting to summarize §4 as “people did not prefer the SPARRING answer.” That summary is wrong about our own instrument. We did not, in the main, ask which finished answer people preferred. Five of the six rubric criteria ask about the *quality of the reasoning that produced the answer*, not the polish of the answer itself:

Criterion	What it probes	Process or product?
C1 Verifiable-artifact citation	does it ground claims in checkable evidence	process — but leaves a fingerprint in the finished text
C2 Substantive vs. theatrical concerns	are the concerns real risks or generic hedges	process — almost literally the “real-or-theater” question
C3 Missed real concerns	catch-rate — does it surface what a careful reader would	process — the work the Challenger exists to do
C4 Calibrated confidence	is stated confidence matched to the evidence	process — partly visible in the text
C5a Factual accuracy vs. briefing	is the answer accurate to its inputs	product
C5b Engagement-with-source	does it grapple with the briefing or skim past it	process — partly visible in the text

Only **C5a** is a property of the finished answer alone. The other five are properties of *how the answer was reached* — and even the head-to-head pass (§4.2) was run per-criterion on these same axes, not as a single “which do you like better.” There was, in effect, no naive output-preference question anywhere in the study.

Here is the contradiction at the center of the design. Those process criteria need to *see the process* — and we deliberately removed it. To hold the blind, both conditions were flattened into the *same* four-section format (§3.2), which equalizes precisely the surface a rater would use to judge *how* an answer was reached. C1 and C5a survive that flattening, because citations and accuracy leave fingerprints in the finished text. **C2 and C3 do not:** you cannot tell whether a concern was *substantive or theatrical*, or *what was caught versus missed*, once the adversarial trace is gone and both answers wear the same shape. We built an instrument to detect substance versus theater — and then removed the stage on which the difference would have been visible. C2, “substantive vs. theatrical concerns,” is almost word-for-word the question of whether SPARRING’s pushback is real or performed; we asked it of raters who could see neither.

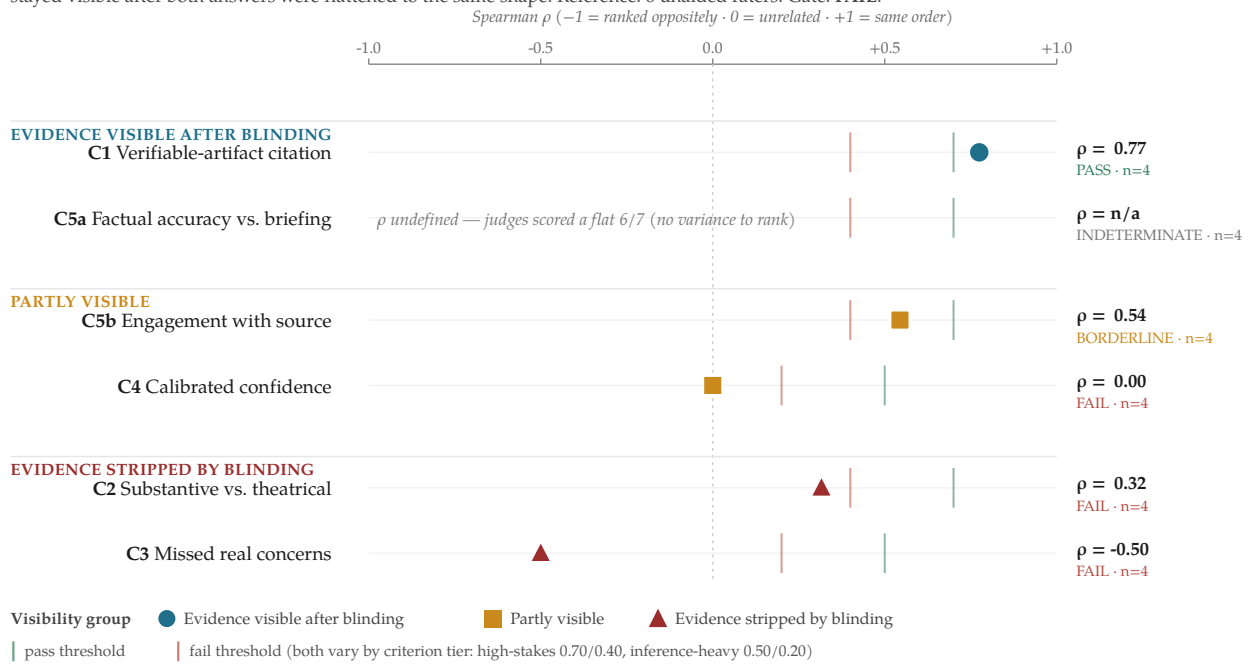
This is not “we asked the wrong question.” On five of six axes we asked the *right* one — process quality. The flaw is an **evidence-instrument mismatch**: process-quality criteria scored against process-stripped evidence.

And our own results carry the fingerprint of that mismatch. Look at where judge-human agreement held and where it collapsed (§4.1). On **C1** — the one process criterion whose evidence survives in the finished text — judges and humans tracked together ($\rho = 0.78$, the study’s only clear pass). On **C2 and C3** — the criteria whose evidence the blind removed — agreement fell to $\rho = 0.32$ and -0.50 . *Agreement held exactly where the evidence was visible and collapsed exactly where it was stripped.* This is not proof — it is one alignment statistic at small n , and C4 (partly visible) failed too — but it is a checkable pattern in the frozen data pointing at *missing evidence*, not merely noisy people, as a driver of the null. The near-floor inter-rater agreement of §4.4 is what you would expect when raters are asked to score a process they cannot see: with nothing to converge on, they scatter.

In plain terms (why this isn’t just wishful thinking). When the author fails the standard test and then argues the test was wrong, you should suspect motivated reasoning. Here is the check that does not rely on our say-so: of the six things raters scored, only

AI judges and people agreed where the evidence stayed visible — and split where it was hidden

Judge–human rank agreement (Spearman ρ) by rubric criterion, grouped by whether the process evidence stayed visible after both answers were flattened to the same shape. Reference: 6 unaided raters. Gate: **FAIL**.



Directional, small-n: each ρ is computed on only $n=4$ paired points — small-sample rank correlations are highly unstable (§4.5).

The pattern is directional, not clean: C4 (partly visible) also fails. Source: `analysis/primary-results.json` · `gate_primary_unaided`.

Figure 2: How often the AI judges and the human raters ranked the two answers in the same order (Spearman ρ : 1 = identical order, 0 = unrelated), criterion by criterion — split by whether the tell that an answer came from the SPARRING *process* was still visible after both answers were flattened to look identical. Where that evidence stayed visible, judges and people agreed (C1); where blinding hid it, agreement collapsed (C2, C3) — a directional pattern, not clean (C4, only partly visible, also fails). Each ρ rests on just 4 paired cases against 6 raters, so the exact values are unstable. (Criteria are defined in §3.3; the per-tier thresholds and the undefined C5a in §3.4.) Source: `analysis/primary-results.json`.

one — *does the answer cite checkable sources?* — still leaves a visible trace once both answers are flattened to look alike. That is exactly the one where the AI judges and the people agreed (the study’s only clear pass). The two that depend on *watching the reasoning happen* — is a concern real or for show, and what got missed — are exactly where their agreement fell apart. So the result lines up with **what the blind left visible, not with what we hoped to find**. That pattern is sitting in the frozen data, anyone can check it, and it is why the null *redirects* the inquiry instead of ending it.

5.2 The V1 reversal, reread — and what we retract

Our V1 pilot found raters *did* prefer the SPARRING output; V2, tighter in every way, found they did not. §5.1 shows why that reversal is close to built-in. The format-symmetric blinding was not only the removal of a *presentation confound* — the elaborate-looking answer made plain — it was the removal of the *process trace itself*. Those are the same act seen from two sides: what made the SPARRING answer recognizable in V1 (its visible deliberation) is exactly the evidence C2 and C3 needed in V2. **Presentation here was never merely aesthetic; it was the audit trail**. Flattening it to protect the blind did not strip noise off a signal — it removed signal and noise together.

That forces a retraction we should make plainly, rather than bank V1 as a win and explain V2 away as noise. V1’s pro-SPARRING result is at least as consistent with “raters could see the deliberation and rewarded it” as with “SPARRING produced a better answer” — the same confound running the other way. The honest reading is that **neither study, as built, could test the claim SPARRING actually makes**: V1 let raters see the process but could not separate its substance from its appearance; V2 separated appearance but, in doing so, blinded the substance.

Internally consistent / externally unvalidated, but worth stating plainly: what survives both is a methodological warning, and it is among the most defensible contributions of this study because it rests on the design logic, not on either study’s noisy headline. **Naive output-comparison of a deliberation method is trapped**. Show the process and preference tracks appearance (V1); hide the process and the rubric’s process criteria go dark (V2). No single blinded output-comparison escapes that dilemma. The dilemma, not either result, is the finding — and it is what sends the inquiry to the process itself.

5.3 Why a blinded output-preference cannot test a process method — and where that leaves us

The honest boundary first. Nothing in §4 *proves* output preference is the wrong test. With near-floor inter-rater agreement (§4.4) this run was uninformative — it could not resolve a difference in either direction — so what follows is an **argument the study motivates, not a result it demonstrates**. We rest the claim on reasoning (and on the §5.1 mismatch) and mark where the data stop.

With that stated: the study put a stranger in front of a *finished answer* to a decision they have no stake in, where **both** sides were written by the same strong frontier model given the same fact-checked briefing. Under those conditions the two final answers are *plausibly* both near a high ceiling, so we would *expect* the visible gap to be small. (That is an inference from the design, not something this study measured — we had no external absolute-quality standard.) If it holds, a small visible gap is not SPARRING failing; it is the test being blind, by construction, to where a deliberation method would do its work.

SPARRING’s claimed value is not “the final memo reads better to an outsider.” It is process value: **what the Challenger catches, and what the decision changes** as a result. A blinded preference over

the polished end-product washes that out, because by the end both answers are decent regardless of how they got there. **Output preference is not the same as process value.** We measured the former and found nothing it could resolve; the latter is the hypothesis worth testing.

This is the pivot the rest of the paper turns on. If the value is not in the *output* but in the *reasoning that produced it*, then the question to test is no longer “is the answer better?” but “**is the reasoning good governance?**” — and that question decomposes into three checkable parts: is the process *consequential* (does it change decisions or catch real defects, §6), is its critique *grounded* rather than confident-sounding theater (§7), and can that grounding be *checked* without simply trusting another AI (§8). The output-preference null does not answer those questions. It is what made us ask them.

6. Governance evidence I — the process is consequential

The first governance property is the bluntest: does the second agent *do anything that matters*? A method can be perfectly auditable and still worthless if all it ever produces is well-cited agreement. So before any claim about grounding, the question is whether the Challenger is **load-bearing** — whether it changes the converged recommendation or catches a specific error the opening pass missed, as opposed to producing well-dressed theater. We lead with this because the failure mode of any pure-transparency story is the shrug: *auditable theater is still theater*. The answer to that shrug is real defects that would have shipped.

6.1 The would-have-shipped catches (verifiable, lead with these)

The strongest evidence is not a rate; it is a short list of specific, externally-checkable catches on *real* engineering and product decisions (not the study’s synthetic cases), each documented in the mechanism-audit corpus ([docs/spars/mechanism-audit-2026-06-26/](#)) and confirmable by a reader without trusting any AI:

1. **An authorization stub that failed open.** On an auth-standardization decision, the Challenger caught that the minimal tier’s identity stubs ran `auth=open|none` and therefore **failed OPEN** — unresolved identities silently authorized — a secure-defaults inversion that would have shipped under the proposed flat opt-in. It reversed the recommendation to a mandatory *fail-closed* identity contract.
2. **An account-lockout rule that was a denial-of-service trap.** On a security-baseline decision, the Challenger reversed a proposed hard per-account lockout as a self-inflicted denial-of-service hazard (CWE-307 — MITRE’s public *Common Weakness Enumeration* catalog of software-flaw types — *Improper Restriction of Excessive Authentication Attempts*; its flip side is that an aggressive lockout combined with low-entropy cohort usernames lets anyone lock out real users), replacing it with throttle + backoff, and added a break-glass recovery path for the sole-admin lockout case the opening missed (CWE-640, *Weak Password Recovery Mechanism for Forgotten Password*).
3. **A scope explosion smuggled in as an “additive field.”** On a runtime-architecture decision, the Challenger flagged that a change framed as a trivial “additive field” was quietly pulling in a much larger messaging system than advertised — the full enterprise request-reply pattern (return address, correlation id, dead-letter channel, idempotent receiver).
4. **A design corrected by historical evidence.** On a motion-accuracy modeling decision, the Challenger pushed back with dated historical sources that *trained* mounted archers still degrade at gallop — forcing a change to the curve-shape design.

5. **False claims about our own product, caught before launch.** On landing-page copy, the Challenger refused to ship marketing claims that the system “builds the page” (it is hand-written) and that personas are a “standing crew that coordinate across projects” (they are invoked in-session, with no autonomous runtime).

These are five distinct, strict-verified case studies (3 from one infrastructure project, 1 product, 1 marketing). The lockout case alone names two real CWEs (CWE-307, CWE-640) a reader can confirm against the MITRE registry as serious, real patterns — but note the boundary of what that verifies: the *CWE is real* is reader-checkable; that the *Challenger caught it in a live ceremony that would otherwise have shipped* rests on the audit corpus (LLM-coded — §6.2). This is the load-bearing evidence for “consequential”: not that the output scored higher, but that a specific, checkable defect did not ship *because* the second agent was there.

6.2 The base rate, adversarially verified — and its hard caveat

To put a number on it, we swept every registered project’s git history and every Claude Code session transcript — 47 candidate blocks, narrowed to **35 genuine ceremony candidates** after dropping non-ceremony material (skill definitions, pre-registrations, reference docs, chatlogs). Coding each then split those 35 into **24 real ceremonies** and 11 that were not actually ceremonies (single-turn prompts, status notes, design specs). A first lenient pass called **23 of those 24** load-bearing (96%) — a near-universal rate that is itself a red flag for coder leniency. So a second, independent pass — one skeptical judge per ceremony, 24 in all — re-scored every ceremony **refute-by-default** against a strict reading of the bar. That adversarial pass downgraded 7 and landed at **16 of 24 (67%)**. The downgrades are the verifier working as intended: it cut elaborately-cited residual-gap lists that changed nothing, SPARK-phase wording polish, and — tellingly — two cases where the converged call was identical to the Generator’s opening (one of which is a V2 study case, consistent with the study’s own finding that the case’s core call did not move).

The hard caveat bounds the claim. This corpus is the ceremonies that left a saved artifact or a rich transcript — i.e. the *hard, contested* decisions people bother to record. The trivial spars that did not matter were never saved. So 67% is “**among ceremonies substantial enough to leave a trace, the Challenger was load-bearing in two-thirds**” — *not* “two-thirds of all spars matter.” This is survivorship bias, and it is the dominant limitation on the number; the adversarial re-judge corrects *leniency*, it cannot correct *survivorship*. The number is also small-n (24) and LLM-coded. The defensible sentence is the conditional one, and we hold to it. The unbiased all-ceremonies base rate is a thing only the prospective ledger (§12) can measure.

6.3 The controlled catch-rate — measured on the producer’s own blind spots

§6.1 is real but unmeasured; §6.2 is a rate but survivorship-biased. §6.2 argued that only a forward-looking ledger could supply an unbiased field rate — but survivorship has a second remedy that does not wait for that ledger to fill: a controlled study that measures *every* recommendation it produces and discards none. It does not recover §6.2’s field base rate; it answers a different, narrower question — on hard problems built for the purpose, how many of a producer’s **own** flaws does each kind of review catch? — and in doing so it trades survivorship bias for a *selection* bias (the problems are chosen, not sampled). We ran one ([pilots/catch-rate-v3-2026-06-28/](#), pre-registered, tag `catchrate-v3-prereg-2026-06-28`), and state that trade in full below.

The design isolates the one thing anecdotes and a field rate cannot: *the producer’s own blind spot*. An agent **produces** a recommendation on a genuinely hard problem in normal mode — the single

pass a user would receive and ship. Three reviews then run on that recommendation, each blind to the others: the **same** agent reviewing its **own** output (self-review), a **fresh neutral** agent, and a **Challenger** (fresh, adversarial, told to work from a different angle). Separately, an **independent 3-agent audit panel**, blind to all three reviews, enumerates the *material* flaws in the recommendation — ones that would change the decision or ship a real defect, not nitpicks — and a flaw counts only if at least 2 of the 3 auditors raise it. That panel-confirmed set is the denominator; each review is then blind-coded for which of those flaws it caught.

The distinction that does the work here is **role, not model**. *Self-review* is the producing agent re-reading its **own** output — same model, same context, judging work it is already committed to, so it carries whatever blind spots shaped the original. A **Challenger** is a *fresh* instance told to attack from a different angle — it did not write the recommendation and has no stake in defending it. When that Challenger runs on the **same API as the producer** (Claude critiquing Claude — the grid below’s *same-vendor* Challenger), the *only* thing separating it from self-review is that fresh, non-author, adversarial stance — the underlying model is identical. So **self-review vs. same-API Challenger isolates the blind spot itself**: hold the model fixed, and change only whether the reviewer is the author. (A *cross-vendor* Challenger adds a second change — a different model — which the grid below separates out.)

On 8 hard problems the panel confirmed **31 material would-ship flaws** (≈ 4 per recommendation; the panel was free to find none, and did not):

Condition	Catch-rate of the producer’s own would-ship flaws
<i>Produce-and-ship — no review</i>	0% — <i>definitional anchor: every flaw reaches the decision</i>
Neutral fresh review	54.8%
Self-review (the producer checks its own work)	67.7%
Challenger	87.1%

(The 0% row is arithmetic, not a measured result: with no review, every audited flaw stays in by construction. Its only empirical content is the ~ 4 -flaws-per-recommendation fact above — and the panel could have returned none.)

The load-bearing comparison is **self-review vs. Challenger**, because both critique the identical recommendation — isolating produce-mode self-blindness from everything else. A producer re-reading its own work leaves **roughly a third of its own material flaws unaddressed (32%)** — the cost of grading your own paper; the Challenger recovers most of them (87%). Head to head, the Challenger caught 8 flaws self-review missed while self-review caught only 2 the Challenger missed — the blind-spot asymmetry SPARRING is built to exploit, measured rather than asserted. And the lift is specifically *adversarial*, not merely a second reader: the Challenger’s 87% far outruns a neutral fresh review’s 55% — the attack posture is doing the work, not just fresh eyes.

We then ran every combination — each of the two models (Claude, GPT-5.2) taking each of the three jobs in turn: producing the answer, challenging it, and auditing it. That full grid (a $2 \times 2 \times 2$ factorial design) separates three things a single-vendor run blurs together:

- **The catch value is robust.** A challenge recovers **78–94%** of the producer’s flaws in *all 8 cells*,

whichever model produces or audits — so “a challenger recovers most of what an unreviewed pass leaves in” does not hinge on vendor.

- **The blind spot is generator-dependent, and real.** Self-review’s miss is large for Claude-produced work but nearly vanishes for GPT-5.2, which self-critiques almost as well as it is challenged. The Claude miss *persists even when a different-vendor panel sets the ground truth* — so it is a genuine blind spot in *Claude-produced* work (on this set), not same-vendor favoritism. Whether a separate challenger beats self-review is a property of the *producing* model.
- **A different-model challenger helps most where the producer is blindest** — best-or-tied in 3 of the 4 generator×audit cells, slightly worse in 1. “Use a challenger, preferably a different model” is a usually-as-good-or-better default, not a universal law.

(Note on the figure below: the **87.1%** Challenger in the table above is the original single-vendor run — Claude challenges Claude, 27 of 31 flaws. The 2×2×2 grid later re-ran that cell, scoring its same-vendor challenger at **90%** (28/31) and adding a cross-vendor challenger at **94%** (29/31). The 87→90 difference is a single flaw — two challenger passes on the same problems, not a pooling. The per-flaw head-to-head below (8 caught / 2 missed, union) is from the original deep run.)

The trade: this study removes the survivorship bias that bounds §6.2 — every recommendation is measured, none discarded — but buys a *selection* bias in return: the 8 problems were authored to a difficulty bar, not sampled, and chosen to be the kind where a strong producer embeds flaws. So “every recommendation carried flaws” is partly a property of that choice, and **nothing here supports a base-rate claim about how often AI is wrong in general**. Three further caveats bound the absolute rates: the same vendor authored, solved, *and* audited the test; the denominator is model-established (a ≥ 2 -of-3 panel, not a human expert — the auditors raised 44 candidate flaws and the consensus bar kept 31, dropping 13 that only one auditor raised, so the denominator is not one model’s opinion); and *n* is small (31 flaws across 8 problems) — enough to show a clear *direction*, not to pin down the exact size, with no statistical test run to rule out chance. Read these as a strong signal, not a settled measurement.

Crucially, those biases mostly fall on the load-bearing comparison equally — with one exception we bound directly. Selection, authorship, and the materiality bar all act on the *shared* flaw set, so they wash out of a within-problem self-vs-Challenger contrast: both conditions are scored against the same flaws on the same problems. The one bias that could *differentially* favor the Challenger is **vendor alignment** — when the Challenger and the audit panel are the same model, they may agree on what counts as material. The grid bounds it: scoring a GPT Challenger against a Claude panel (and the reverse) shows alignment is worth only ~2–5 points, and the self-vs-Challenger asymmetry survives the cross-vendor audit. Human-expert ground truth and an independent-source problem set are the named upgrades (§12).

Together §6.1–§6.3 triangulate one property from three directions, each bounded differently — real-but-unmeasured anecdotes, a survivorship-bounded field rate, and a selection-bounded controlled measure. They converge on the robust claim: a challenge recovers material flaws that an unreviewed pass would ship — the second agent is **consequential**. Whether a *separate* challenger is needed beyond the producer’s own self-review is the stronger, generator-dependent claim that only §6.3 isolates (large for Claude, marginal for GPT-5.2) — established here, not by the anecdotes or the field rate.

A challenge recovers 78–94% of flaws in every model × role combination

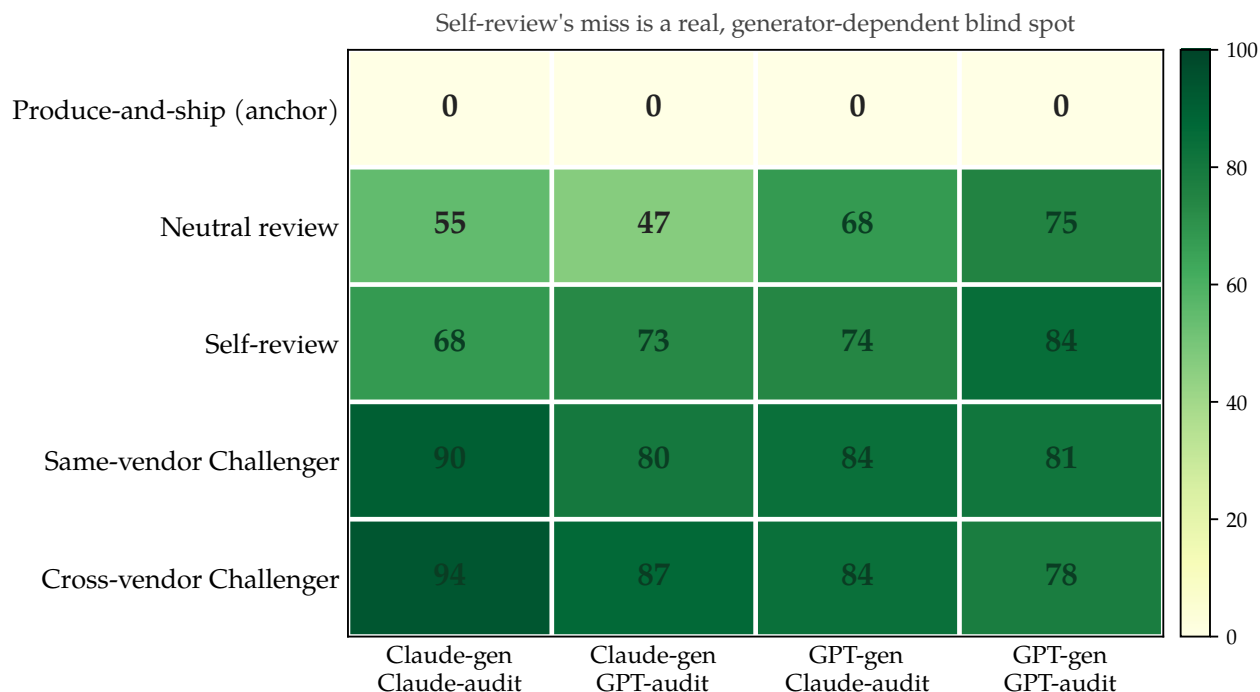


Figure 3: Catch-rate of the producer's own would-ship flaws across every combination of the two models in three roles — producing, challenging, auditing (a 2×2×2 grid). A challenge recovers 78–94% of flaws in all eight cells regardless of which model produces or audits; self-review's miss is large for Claude-produced work and nearly vanishes for GPT-5.2, and persists for Claude even when a different-vendor panel sets ground truth — a genuine generator blind spot, not same-vendor favoritism. The produce-and-ship row (0) is a definitional anchor, not a measured cell. $n \approx 30\text{--}32$ flaws per cell; model-established ground truth ($\geq 2\text{-of-3}$ panel); blind-coded by a single held-constant coder; directional pilot, no significance test. Source: results-2x2-grid.json.

6.4 The both-must-agree gate’s controlled test: iterate-to-agreement is an output-quality null

§6.3 isolated a *single* adversarial challenge. SPARRING’s both-must-agree gate (§2), though, would drive that challenge *toward* a *revised, agreed output* — and §6.3’s dependent variable (flaws *flagged* on a fixed recommendation) cannot see it: it ceilings at 87% and has no revision step to score. This completing study isolates the gate on a dependent variable that can: the material would-ship flaws *remaining in the final output*, counting the new flaws the revision introduces. Pre-registered before any data (resolver-iteration-prereg-2026-06-29).

Four conditions produce a final deliverable, all scored identically by the §6.3 audit panel against the same ≥ 2 -of-3 panel-confirmed flaw set on the same eight problems: **R0** ship-as-is (anchor), **R1** one challenge + one revision, **R2** iterate-to-agreement (the gate: challenger $\leftrightarrow$ producer until both agree, stall, or a hard cap of five rounds), and **R1’** an effort-matched control (the producer revises twice on its own, no challenger). Two vendor arms: same-vendor (Claude challenger) and cross-vendor (a GPT-5.2 challenger against the Claude producer — the sharpest test, since same-vendor agents are the likeliest to fake agreement).

The load-bearing contrast, R2 vs R1, is a null. Iterating to agreement leaves no fewer net material flaws *per output* than a single challenge-and-fix: a dead-even **+0.000** average across the eight problems same-vendor (2 better, 2 worse, 4 tied), **−0.125** cross-vendor (3/3/2). Nor does the gate beat the effort-matched self-revision control: R2 is *worse* by **+0.375 flaws per output** same-vendor and **+0.750** cross-vendor. The adversarial back-and-forth is not what yields a cleaner output here; on this measure it slightly hurts.

The mechanism is the part worth reading, and it is consistent with the rest of the report. The challenge is a genuinely better *detector* — challenged conditions remove **87.1%** of the producer’s original blind-spot flaws (0.5 remaining of the 3.875 per problem — §6.3’s 31 panel-confirmed flaws spread over the eight problems), numerically identical to §6.3’s single-vendor catch-rate (27/31), reproduced here through a different instrument (re-auditing the revised output rather than coding flags). But revising *under adversarial pressure* seeds **new** material flaws (~ 2.0 – 2.6 per output) that quiet self-revision largely avoids (~ 1.25) — a between-condition difference that is the load-bearing part; the absolute counts include a fresh audit panel’s baseline propensity to surface issues on any output, so read the levels as upper bounds. The detection win and the churn cost cancel, and once new flaws are counted — and compared against an effort-matched control — the gate’s marginal advantage disappears. (Without the new-flaw accounting this study would have reported a headline-friendly 87% “catch” and been wrong about it.)

One pathology the gate exists to prevent did **not** appear: fake agreement — agents agreeing to stop while flaws remain. Of sixteen ceremonies, four converged, twelve hit the cap, none stalled; the converged outputs were *cleaner* (1.25 net flaws) than the capped ones (3.33), the opposite of a convergence-gaming signal. This is an *absence of signal at $n=8$* , confounded with difficulty (agreement was reachable on the cleaner problems), not proof the gate cannot be gamed — but the sycophancy the gate is built to stop did not surface, including in the cross-vendor arm where it was predicted to bite hardest (which converged once, on a clean net-zero output).

What this means for the thesis. The gate’s iterate-to-agreement mechanism does not produce a higher-quality converged answer than a single challenge — a *null on output quality*, pre-committed and reported as a finding. Far from undercutting the report, it is the same lesson the output-preference study taught (§4–§5), now at the gate itself: the value of SPARRING’s Resolver is not that more adversarial iteration yields a better *answer*. That null also rhymes with the broader litera-

Iterating to agreement buys no cleaner answer than a single challenge

— it removes blind spots, but revising adds new flaws, and the two cancel (R2 vs R1 = the null; §6.4)

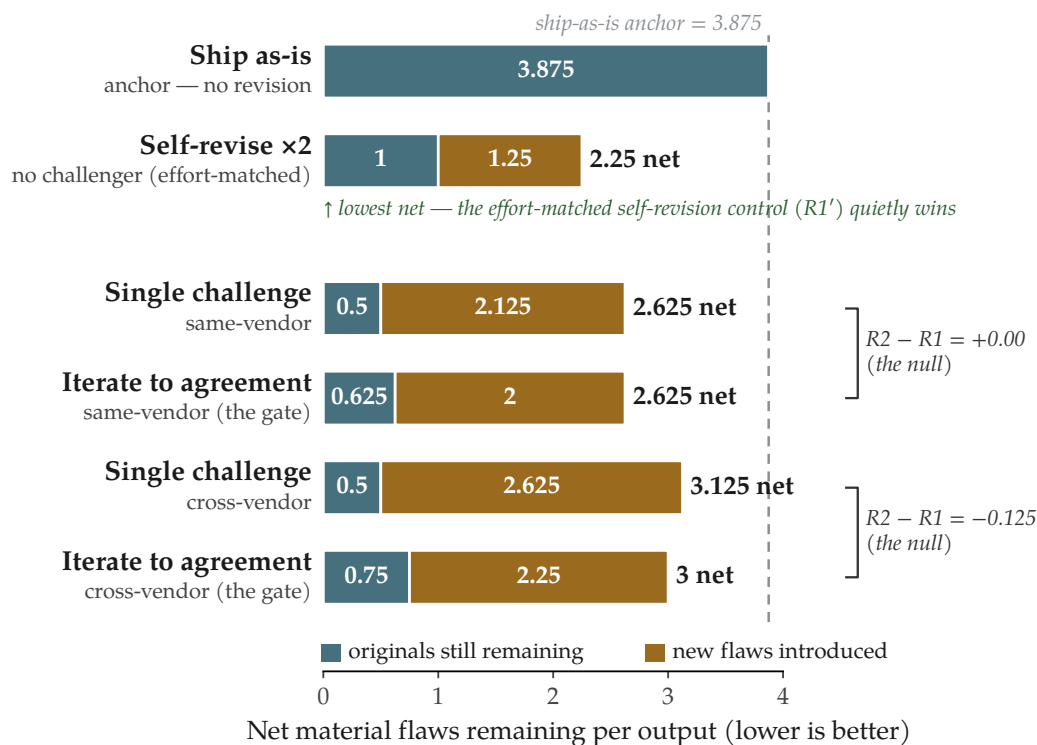


Figure 4: Net material would-ship flaws remaining in the final output for each condition, split into the producer's *original* blind-spot flaws still present (teal) and *new* flaws the revision introduced (amber) — lower is better, against a ship-as-is anchor of 3.875 flaws per problem. Any challenge is a strong *detector*: the teal segment collapses to ~0.5, about 87% of originals removed, matching §6.3's 27/31. But revising under adversarial pressure seeds ~2.0–2.6 new flaws where a quiet self-revision (R1') seeds only ~1.25, so the two effects cancel and the net bars barely move. Iterating to agreement (R2) versus a single challenge (R1) is the pre-committed null — +0.00 same-vendor, −0.125 cross-vendor — and the effort-matched self-revision control has the lowest net of all. Directional; n=8 paired problems; model-established ground truth (≥ 2 -of-3 audit panel), no significance test; absolute new-flaw levels include a fresh panel's baseline propensity to flag any output, so read them as upper bounds. Source: resolver-iteration-2026-06-29 results.md / analysis/results-aggregate.json.

ture: models do not reliably self-improve by iterating on their *own* judgment without new external signal (Huang et al. 2024; Stechly, Valmeekam, and Kambhampati 2024), while the *single* external challenge of §6.3 — which recovered real would-ship flaws — matches the pattern where added external feedback, not self-directed revision, is what does the corrective work (Shinn et al. 2023; Gou et al. 2024); reported gains from self-refinement (Madaan et al. 2023) concern more open-ended tasks than the fixed-decision setting here. A structurally similar trade shows up one level down, in the judge rather than the ceremony: asking an LLM reviewer for an explanation and a proposed fix, not just a verdict, cuts false acceptance of bad code but drives false rejection of *correct* code from roughly 26–36% to 73–88% on two of three coding benchmarks tested, across five models (Jin and Chen 2026) — the same detection-improves, rejection-worsens shape this section finds in the gate’s own mechanism, in an unrelated domain (code requirement conformance, not adversarial revision). The Resolver’s value, then, is the governance properties established in §6.1–§6.3 and §7–§8. The gate earns its place *by design* — §2’s non-bypassable stop makes faked agreement structurally hard — not as a quality pump; this study adds only that iterating it buys no cleaner answer, and, at $n=8$, surfaces no fake-agreement signal (absence of evidence, not proof the gate prevents gaming). (*Directional, $n=8$, no inferential statistics; the pre-registered expansion trigger to ~16 problems is met and noted as future work — it requires authoring new problems and their ground-truth denominators, not a re-run.*)

In plain terms. We tested SPARRING’s back-and-forth move — make the two agents argue until they agree — against just challenging once. Arguing to agreement did not produce a cleaner final answer: the challenge is good at *finding* the author’s blind spots, but rewriting under pressure introduces about as many new problems as it fixes, and a second quiet self-edit does just as well. That is not a knock on the method; it is the same finding as the rest of the paper — the payoff is catching consequential, checkable problems (and being able to verify the catch), not squeezing out a better-scoring answer.

6.5 Can a gate *learn* to block regressions? In this exploratory probe, an LLM admissions gate did not discriminate them

§6.4 retired the iterate-to-agreement gate on quality grounds. A natural repair suggests itself: rather than iterate a challenge *toward agreement*, **automate a gate** that inspects a proposed revision and *blocks the ones that made the output worse* — an admissions gate on the revision itself. If such a gate worked, the machinery §6.4 found inert would be replaced by something that earns its keep. This exploratory arc built and measured that gate, and it did not work — in a way that is itself informative.

The test set is the §6.4 study’s own **61 champion→candidate revision decisions**, each labeled by the same net-flaws-remaining ground truth: a revision is a *regression* if its net material flaws rose (11 of 61). Across five successive gate designs — a lenient rubric gate, a budgeted variant, a surgical one, the §6.3 challenge-move recast as a gate, and a purpose-recrafted zero-shot gate — the gate’s ability to *separate* a real regression from a clean revision never meaningfully cleared chance:

gate	recall (11 regressions)	precision	balanced acc	Cohen’s κ	block rate
static verifier	0.27	0.38	0.586	0.193	0.131

gate	recall (11 regressions)	precision	balanced acc	Cohen’s κ	block rate
challenge-move gate	1.00	0.19	0.520	0.015	0.967
recrafted gate (zero-shot)	0.00	—	0.500	0.000	0.000

(The table shows the original static verifier and the two extremes of the recraft sweep — the block-94% challenge-move and the admit-100% recraft; the lenient, budgeted, and surgical variants fall in the same 0.50–0.59 discrimination band. Full per-gate numbers: `r6-gate-findings.md`.)

The three gates span block rates from 0% to 97% — yet all three sit at **balanced accuracy ≈ 0.50 –0.59**, $\kappa \approx 0.00$ –0.19. Prompt framing is a blunt *rate* control: it slides the operating point along the precision/recall axis (admit-everything to block-everything) without improving the gate’s underlying *discrimination*. The wall is discrimination, not calibration — consistent with the pattern that labeled examples, not prose instructions, are what calibrate an LLM judge (these gates ran zero-shot). The best of everything tried remains the original static verifier at $\kappa \approx 0.19$, which still catches only 27% of regressions.

Before writing that as a limit of LLM gating, one confound had to be ruled out: **is the ceiling the model, or the task?** Every gate above ran on the same substrate (`claude-opus-4-8` + `gpt-5.2`). So we re-ran the frozen recrafted gate — same 61 candidates, same prompt, same ground truth, same panel *composition* — with the two Claude panel members upgraded to the most capable model available at writing (`claude-fable-5`), two ways: the mixed panel (2 Fable + 1 GPT, holding vendor diversity constant), and a 3× Fable solo panel (removing GPT as a possible drag on the vote). Both are flat against the opus baseline — balanced accuracy **0.500**, κ **0.000**, every one of the 61 candidates admitted, all 11 regressions missed — with **zero refusal or fallback contamination**: no member fell back to opus, and the frontier model simply never assigned a nonzero defect severity to any candidate. The frontier tier moves nothing. **A better model does not repair it** — capability is not the missing lever.

What this establishes, and what it does not. It establishes — *exploratory, not pre-registered* — that on this evidence an LLM admissions gate’s *block rate* is a wording artifact while its *discrimination* of material regressions is not, and that the failure is not repaired by the most capable model available, tested two ways — echoing an independent finding, on an unrelated classification task (code-requirement conformance, not revision admissions), that asking an LLM judge for reasoning and a proposed fix shifts its block rate without a matched gain in discrimination, trading detection for over-rejection (Jin and Chen 2026). It does **not** yet establish that LLM gating *cannot* discriminate regressions, because a second confound is still open: **the labels themselves may be noise** (each row’s label is its regression / not-regression verdict — the answer key the gate is graded against). Only 11 of 61 rows are regressions, defined by a ± 1 -noisy net-flaw score, so some of those 11 are scoring jitter rather than real regressions — and no gate can discriminate a label that is itself noise. A flat frontier-model result on noisy labels cannot separate “no model can discriminate” from “there is nothing here to discriminate.” Closing that gap has two steps, both carried as future work (§12): replicate-score the labels, and — if the de-noised regressions prove real — try the one lever these runs did not, labeled few-shots mined from gate-vs-truth disagreements. The honest reading today is a **forming** negative result: the *capability* confound is closed, the *label-noise* confound is open, and the few-shot lever is untested. On current evidence the direction reinforces this report’s the-

sis — the machinery that would sit *on top of* the single grounded challenge, iterate-to-agreement (§6.4) and now an automated admissions gate, has not *yet* earned its keep, and the validated kernel remains **one challenge, apply, stop** (§6.3). (Source: [pilots/resolver-iteration-2026-06-29/analysis/r6-gate.json](#), [r7-fable-probe.json](#), [r7-fable-solo.json](#); findings in [r6-gate-findings.md](#), [r7-fable-probe-findings.md](#).)

In plain terms. After finding that arguing to agreement didn’t help (§6.4), we tried the other obvious upgrade: an automatic gate that catches the revisions that got *worse* and blocks them. Across five versions, the gate could be reworded to block anything from *never* to *always* — but it never got meaningfully better at telling a genuine regression from a clean edit, and swapping in the most capable model available didn’t change that. One caveat we’re still chasing: the “worse” labels are themselves a bit noisy, so part of what looks like an unbeatable wall may just be the labels — we’ll know once we clean them up. The takeaway stands with the rest of the paper: the extra machinery stacked on top of a single grounded challenge keeps not paying for itself.

7. Governance evidence II — the critique is grounded, not theater

Consequential catches answer “does it matter.” The next governance property is harder, and two questions hide inside it. The **absolute** one: is the second agent’s concern *grounded* — backed by a checkable artifact — or is it confident-sounding *theater*? And the sharper **comparative** one: does that grounding actually *distinguish* SPARRING from a competent skeptic who is merely arguing harder? This is the real-or-theater question the blinded study (C2) was built to ask and structurally could not see. So we built a study that could: a **seeded-defect pilot** that measures grounding directly, on the critique, with the output-preference question removed ([pilots/seeded-defect-2026-06-26/](#), pre-registered, tag [seeded-defect-prereg-v3-2026-06-28](#)) — the **v3 run**, with grounding adjudicated mechanically rather than by one model judging another.

7.1 Design

Twelve LLM-generated flawed recommendations, each carrying one seeded defect from four families that differ in *where the check lives*: misapplied or fabricated **citations** and **security/CWE** flaws (verifiable only against *outside published references* — the public security-flaw registry, the citation record), and **overclaims-against-spec** and **buried contradictions** (verifiable against facts *inside* the briefing the agent was given). Each case ran under three conditions: **B**, a neutral single pass; **S**, the full dual-grounded SPARRING ceremony (Generator and Challenger together); and **T**, a **realistic-skeptic placebo** — a tough reviewer told to push back, free to point at facts in the briefing, but with no artifact-citation discipline and no outside lookup. **T** is the load-bearing control: what “just argue harder” looks like done *competently* (it replaces an earlier no-discipline strawman placebo). Every concern was coded **blind** (condition hidden, order rotated) for two things: is it *legitimate*, and is it *grounded* in a checkable artifact. Grounding was checked **mechanically**: every cited artifact was existence-checked against public registries (CWE/RFC/DOI), every in-briefing fact against the saved briefing (no model in the loop), and then a separate check — run on *two independent vendors* (Claude and OpenAI) — confirmed that each resolved source actually *backs* the claim it was cited for. (We label this machine route **Path-A**, versus the AI-coded route **Path-B** that §8 refers to.) So the grounding numbers are mechanically verified and cross-vendor, not one model rubber-stamping another. 99 concerns were coded (B 12, S 27, T 60).

The pre-registered null worth naming: if $S \approx T$ on grounding, then SPARRING’s artifact discipline

adds nothing a competent skeptic doesn't already do — the apparent rigor would be theater. That is the null this study could return.

7.2 Result: the critique is grounded; the edge over a *competent* skeptic is real but small

A quick key: **grounding rate** = share of concerns backed by a checkable artifact; **precision** = share of flagged concerns that are real, not false alarms; **theater rate** = share that are confident assertion with nothing to check.

Condition	concerns	Grounding rate	Precision	Theater rate
S (ceremony)	27	96.3%	100%	3.7%
T (realistic skeptic)	60	83.3%	100%	16.7%
B (neutral pass)	12	58.3%	50.0%	41.7%

Two findings, the load-bearing one first:

1. **Against a raw single pass (S vs B), the direction is robust — and there is no precision cost.** An adversarial review grounds and targets its concerns far better than an un-adversarial one: S is **96% grounded and 100% precise**, a raw pass **58% / 50%** — and the rigor carries no precision penalty (the feared “more pushback just means more false alarms” did not appear). But read these as *direction, not effect size*. The magnitudes are generation-sensitive — B's 50% precision is largely a few “looks-fine, ship-it” statements coded as false-positive concerns — and because the realistic skeptic raised far more concerns than S (60 vs 27), the conditions are compared as *rates*, by design: on raw counts the verbose skeptic actually surfaced more grounded-legitimate findings than S, so rate-normalization is the fair lens, and the S-over-B direction holds under it.
2. **Against a *competent* skeptic (S vs T), the artifact-discipline edge is marginal.** S grounds 96% versus a realistic skeptic's 83% — a ~13-point edge at $n = 12$, no significance test. The skeptic grounds well precisely because the decisive evidence is often *in the briefing*, where anyone can point at it. So most of SPARRING's rigor advantage over a raw pass comes from **having a genuine challenger at all**; the citation discipline *on top* adds a modest further edge on these easy, evidence-in-hand cases.

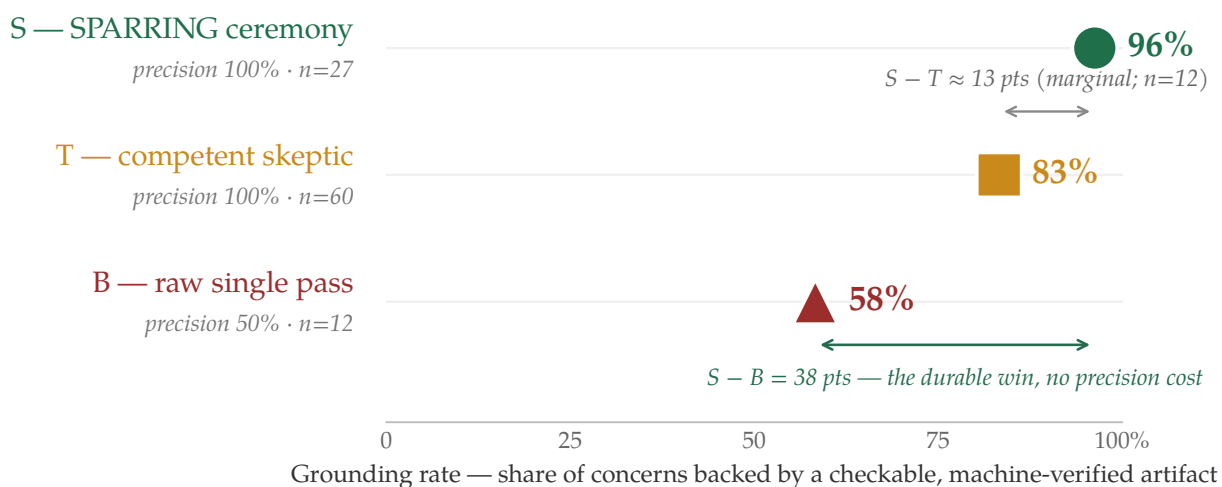
The fragility finding. This run first used a strawman placebo — told to cite nothing — which grounded just **15%**, making S look like a crushing 96-vs-15 win. Regenerated as a *realistic* skeptic, the same control grounds **83%**, and S's edge collapses to the marginal ~13 points above. (It also supersedes the earlier-draft v2 result, whose placebo grounded 60.7% under AI-coded adjudication — a third, intermediate control; the lesson is the same.) The measured “theater gap” is therefore **fragile**: it depends almost entirely on how weak the control is and whether the decisive evidence is in hand. Reporting the 15% would have badly overclaimed; the 83% is the honest number, and the headline is the win over a raw pass (finding 1), not a theater knockout over a competent skeptic. This is itself a methodological result for anyone evaluating deliberation methods: *your placebo's competence sets your effect size*.

7.3 Where the small edge concentrates — directionally, the external-check families

The artifact-discipline edge is not uniform, and its shape is the governance-relevant part. Cells below are *grounded concerns / concerns raised* per condition — read as direction, not point estimates

SPARRING's critique is grounded — the durable win is over a raw single pass, not over a competent skeptic

v3 seeded-defect pilot. Each condition placed by its grounding rate on a common 0–100 scale.
S leads a raw pass (B) by a wide margin at no precision cost; its edge over a competent skeptic (T) is marginal.



Grounding is machine-verified: existence model-free vs. public registries / the saved briefing; correct-application is cross-vendor (Claude+OpenAI) — §8. Precision/legitimacy is LLM-coded and awaits the domain-expert audit. Directional: n=12 cases / 99 concerns, no significance test. Source: v3-run/aggregate.json.

Figure 5: How often each reviewer’s concerns came with a checkable source — its “grounding rate” — in the v3 seeded-defect pilot: a raw single pass (B) 58%, a tough competent skeptic (T) 83%, the SPARRING ceremony (S) 96%. The durable result is S’s large lead over a raw pass at no cost to precision (its flags are just as often real); the edge over a competent skeptic is marginal — about 13 points (12 cases / 99 concerns, no significance test). Grounding is machine-checked — existence against registries with no model, correct-application judged cross-vendor — not a single model grading itself; precision/legitimacy is model-coded and awaits the domain-expert audit. Source: v3-run/aggregate.json.

(each family rests on only 3 cases):

Defect family (where the check lives)	B	S	T
Citations (external: the literature)	1/3	9/9	11/15
Security / CWE (external: MITRE registry)	3/3	6/6	12/15
Overclaim-vs-spec (in the briefing)	0/3	6/6	14/15
Buried contradiction (in the briefing)	3/3	5/6	13/15

Directionally the S-over-T edge is largest on the **external-check** families — citations and security, where grounding a claim means reaching outside the materials — and shrinks to nothing or reverses where the check lives **inside** the briefing (on buried contradiction S even runs *below* the skeptic, 5/6 vs 13/15). That is the governance-relevant direction: the artifact discipline helps most exactly where a reader *cannot* verify by skimming the briefing. Two cautions. Each family rests on just **3 cases** — far too few to trust the magnitudes, and the contradiction reversal is as plausibly noise as signal. And even on the external families the realistic skeptic grounds most of what it raises (12–13 of 15), so this is a *modest concentration of a small edge*.

7.4 What this does and does not show

This shows SPARRING produces **verifiably grounded, high-precision critique** — far more grounded than a raw pass, and modestly more grounded than even a competent skeptic where the evidence is in hand — a governance property, now established under **objective, cross-vendor adjudication** rather than one model judging another. It does **not** show better *outcomes* or better *answers* (the retired output-preference claim, §5). And it pointedly does **not** measure **catch-rate** — whether the challenger surfaces flaws a raw pass would miss: these seeded defects are obvious enough that every condition catches them, and the decisive fact often sits in the briefing, so this design is structurally blind to that question. Catch-rate is a *distinct* property, measured in §6 — the real would-have-shipped catches (§6.1) and the controlled self-vs-challenger rate (§6.3). §6 does not stand in for §7's grounding result, nor this for §6's: they are two different governance properties, each measured by the instrument that can see it.

The grounding result is bounded by caveats carried into §11: $n = 99$ concerns / 12 cases, directional, no significance tests; **legitimacy** (whether a flagged concern is real) still awaits the domain-expert audit (§12, item 3) — only *grounding* is machine-checked here; the ceremony agents share a vendor family on the Claude side (the cross-vendor half of the backs-the-claim check mitigates, does not eliminate); and S costs $\sim 2\times$ (two agents). §8 reports how much of this grounding the verifier resolves mechanically — the coverage question that complements these rates.

8. Governance evidence III — grounding you can check mechanically against external truth

A skeptic’s first question about §7 should be the obvious one: *who checked the grounding?* “An AI says the AI’s reasoning is well-grounded” is exactly the kind of self-certifying loop to distrust — and should be. But §7’s grounding was **not a single model certifying its own reasoning**: whether each cited artifact *exists* was confirmed mechanically against external registries (no model at all), and whether it actually *backs the claim* was judged across **two independent vendors** — so no model ever vouches for its own family’s work. That two-layer design — model-free where it can be, cross-vendor where a model is unavoidable — is the **Path-A** route of §7, and §8 is the instrument: what it checks, how far it reaches, and where a human is still required.

The distinction it earns is **verifiable, not merely transparent**. Transparency — shipping the transcript — is cheap and nearly automatic; a method that simply logs its reasoning has cleared a low bar. The hard, non-obvious part is **explainability you can trust**: telling real reasoning from theater. The verifier tells them apart in two layers (pilots/seeded-defect-2026-06-26/scripts/):

- **Layer one — does the cited source exist? (no API keys, no model.)** Each cited artifact is looked up in a public registry to confirm it is real — **MITRE** for security flaws (CWE), **Crossref/DOI** for citations, the **IETF datatracker** for RFCs (the internet’s published technical standards) — with checks also for code/file references (against the working tree), in-briefing facts (against the supplied briefing), and arbitrary web URLs (a read-only fetch: a live page confirms, a dead or fabricated link is refuted). This catches *fabrication* — a source cited but nonexistent — with no model in the loop, and an unconfirmable artifact is a **durable** state: counted against the grounding rate, never silently upgraded. (Its automated self-tests cover every source type and the live-URL fetch.)
- **Layer two — does the source actually back the claim? (cross-vendor.)** For an artifact that does exist, a second check catches *misapplication* — a real source cited to support something it does not. Per the project’s cite-check house rule it runs on **two independent vendors (Claude + OpenAI)** and surfaces disagreement rather than forcing convergence. Demonstrated on real content: fed a claim that cited Sweller’s cognitive-load theory (1988) *backward*, it returned **REFUTES**, quoting the source against the claim; on the earlier external-canon run it agreed with the (LLM) Path-B coding on 15 of 17 cited artifacts — the two disagreements both tool-side and in *opposite* directions (one over-crediting a rhetorical CWE mention, one a citation-extraction gap), neither a fabrication and neither overturning the coding’s direction.

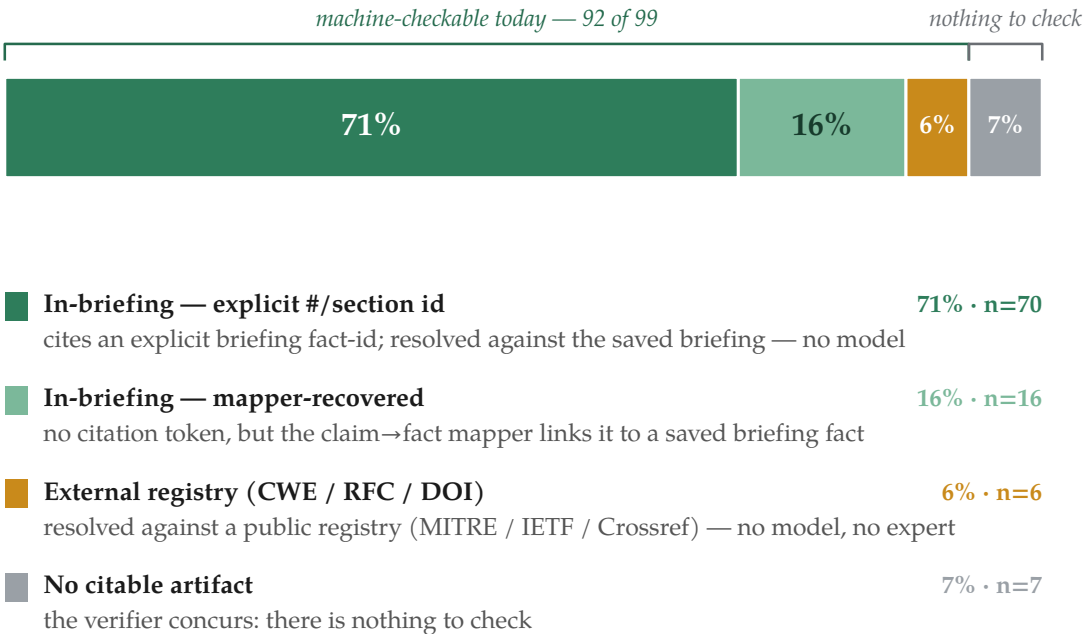
How the check runs, on one concern — no black box (Figure 7; full trace in Appendix C). Take the lockout finding’s cited artifact, CWE-640. A regex *classifier* reads the citation’s **shape** — `CWE-\d+` — and routes it; it never infers the topic, only the citation *type*. The matching backend does one hardcoded lookup — MITRE’s registry API for weakness 640 — with **no model and no web search**: a 404 refutes (a fabricated id), a hit confirms existence. Every backend has that shape — an RFC number → the IETF datatracker, an author-year or DOI → a Crossref *search*, a `path.php:214` → a read of the working tree, a URL → a live fetch, a `#3/$2` → a text-match against the briefing supplied. Only *after* existence is settled does a model enter: the cross-vendor judge (Claude + OpenAI) rules whether the resolved source actually *backs* the claim. A citation matching no backend is counted ungrounded, never waved through. So “no model in the loop” for existence is literal — the fabrication check is a pattern-match plus an API call a reader can run unaided.

The coverage that makes §7 Path-A. The verifier checks the **large majority of concerns mechanically, across all four defect families**: of the 99 concerns in the v3 corpus, **86 are confirmed against**

the saved briefing (directly, or via the claim-to-fact mapper that links a prose claim to the briefing fact it leans on), 6 against external registries (MITRE / Crossref / IETF), and just 7 carry no citable artifact at all ([pilots/seeded-defect-2026-06-26/v3-run/](#)). That coverage is new to v3: in the earlier (v2) corpus only the external-canon slice (~7%) could be machine-checked, because the per-case briefings were never saved — a data-loss artifact, not a tooling gap (the in-briefing and code resolvers already existed). v3 persists every briefing, so they finally have data to run against, and the gap is closed. *Before, a machine could independently confirm almost none of the grounding and the rest rested on one model’s read of another’s; now it confirms the bulk* — which is what lets §7 call its grounding machine-adjudicated rather than AI-graded.

A computer can independently check the grounding of 92 of 99 objections in v3

All 99 v3 seeded-defect concerns, grouped by how their grounding is resolved. v3 persists every briefing, so the in-briefing majority is now machine-checkable — closing v2’s ~7%-only ceiling, a data-loss artifact, not a tooling gap. §8 / §10.2.



Existence is model-free; correct-application is cross-vendor (Claude+OpenAI). The residual limit is now LEGITIMACY (whether a concern is real) — not coverage — which routes to the domain-expert audit, not the verifier. Directional, synthetic; n=99 / 12 cases. Source: v3-run/coverage.json.

Figure 6: How the grounding of all 99 v3 concerns is checked: 86 against the saved briefing (70 cite a tagged briefing fact directly, 16 are linked to one by a tool that maps a prose claim to the fact it leans on), 6 against external public registries (MITRE/Crossref/IETF), and 7 with no citable source — so 92 of 99 can be checked by machine, up from only ~7% in v2, which failed not for lack of tooling but because the per-case briefings hadn’t been saved; v3 saves them all. Existence is checked with no model; correct-application is judged cross-vendor; legitimacy is not checked here and routes to the domain-expert audit. Directional, synthetic; n=99 concerns / 12 cases. Source: v3-run/coverage.json (derived from case-file codings).

What the verifier reaches — and what it doesn’t. The reach has a clean shape across three axes.

Existence is the strong, model-free half: a reader (or an automated check in a build pipeline) can confirm a cited CWE or DOI is real without being a security engineer or a subject specialist — no model, no domain expert. **Correct application** is the harder, model-based half — the cross-vendor judge above. And **legitimacy** — whether a flagged concern is actually *real* — is a third axis the verifier does not touch at all: it routes to the domain-expert audit (§12, item 3). So v3 makes *grounding* objective and cross-vendor across families; the *legitimacy* rates in §7 remain expert-pending (with the broader pilot caveats carried in §11). The verifier is, today, a pilot-local tool; the intent is to make it a standing, reusable step run *at ceremony time*, not only in post-hoc analysis.

Is this security-specific? No — the check is organized by artifact type, not by domain. Most of it needs no external source at all: the 86-of-99 in-briefing majority asks only whether a concern cites a fact present in the materials the agent was *already given* — domain-independent by construction, and the same discipline the two in-briefing defect families (overclaim-vs-spec, buried contradiction; §7.3) exercise. The external registries are **wired backends, one per source type** — MITRE/CWE (security), Crossref/DOI (scholarship of any field), IETF (standards) — alongside a generic live-fetch for any cited URL (built in the productized verifier; stubbed in the copy used for this report, which no case in this corpus needed). Covering a new domain’s *out-of-document* claims means wiring that domain’s own registry as one more backend (engineering done once, not per decision); the verifier does not autonomously discover a canon for an arbitrary claim. Where a domain has no such registry, model-free checking narrows to the in-briefing, code, and URL resolvers, and the remaining out-of-document claims route to the expert legitimacy audit (§12, item 3). So the governance *mechanism* — a concern counts only if it cites something checkable in the given materials — generalizes across domains; what is registry-dependent is the strongest form of the *external* guarantee (fully model-free, no domain expert), not the method.

8.1 Extending the check to the answer’s own figures — the definitiveness gate (exploratory)

Exploratory; not pre-registered. n=12 natural-corpus recommendations (real recommendations, not cases built for the test; resolver-iteration pilot). The detector, calibration, and revision study live in the pilot repo; this is the claim and its load-bearing evidence.

§8’s verifier turns an external eye on the *Challenger’s* cited artifacts. The same discipline turns just as naturally on the *answer’s own* load-bearing figures — a **definitiveness gate** (the check that no stated number is taken as established unless it traces to the given facts or is marked an estimate). It extends the grounding *check* from the slice of claims that cite external canon to the specific quantities a recommendation states — mechanically where traceability to the givens can be computed, cross-vendor where the presentation call needs a model.

The rule is one a reader already applies without thinking: an unmarked figure is read as **definitive**. So each hard specific — a cost, a rate, a duration, a threshold — must be one of:

- **traceable** — it follows from the problem’s given facts (echoed, or derivable from stated numbers), decided by a **calculator, with no model in the loop** — the model-free analogue of §8’s existence layer; or
- **contextualized** — explicitly marked as an estimate, assumption, or example (“assume ~\$X; verify against your actuals”); or
- **neither** — an unmarked figure that does not trace to the givens: an *implicit false-definitive*, and the gate’s failure state.

As in §8, the split is model-free where it can be and model-based only where it must be. Whether a

stated figure reduces to the givens is decided by the calculator on inputs quoted from the problem — the model only *proposes* a derivation, and Python adjudicates it. One judgment genuinely needs a model: is a figure *presented* as fact, or marked as an estimate? That call is made **cross-vendor** (Claude + OpenAI). On a shared rubric the two agree ~90% of the time — agreement, not correctness; there is no gold here, and the figure is calibration-set — and the residual disagreements are **escalated to review**, not silently resolved (the cite-check house rule again: surface splits). The check runs on a **two-part output** — the recommendation, plus a *trust rider* carrying the derivations, the flagged assumptions, and any escalated split — so the accounting sits beside the answer instead of diluting it.

On 12 natural-corpus recommendations the gate surfaces the unsupported figures a careful reader would want caught: a monthly cost advantage stated at under half its derivable value, a per-line saving that cannot be isolated from the numbers provided. (Directional, $n=12$; precision on its intended target — unsupported factual claims — is high: ≈ 9 of 11 descriptive flags genuine at small n , once the recommendation’s own *proposed actions* — a pilot length, a deadline it sets — are set aside, since a choice cannot be “false.”)

A consequential demonstration (constructed positive control — cases built with a known-correct answer, to confirm the check flags a real error). To show the catch is decision-relevant and not cosmetic, three cases were built with **arithmetic ground truth**: each turns the decision on a *single* figure crossing a threshold — a budget cap, cash on hand, a deadline — and each recommendation states that figure *wrong* and, on it, picks the wrong option. In all three the gate flagged the figure as untraceable, and on correction the decision flipped to the option consistent with the binding arithmetic — the *arithmetically*-correct choice, not a “better” or preferred one. In one, an aggressive-hire plan is claimed to cost “~\$650,000, within the \$700,000 cash on hand,” but the givens yield $8 \text{ hires} \times \$12,000/\text{month} \times 9 \text{ months} = \$864,000$, over cash — corrected, the choice flips to the affordable plan. This is a **consequential catch in the §6.1 sense** — a wrong figure that would have shipped a wrong *decision* — but established on constructed, single-threshold cases with a calculator as judge, so it is a positive control for the instrument, not a base rate.

The framing is the paper’s: a *consequential, externally-checkable* catch (§6.1’s kind), **not** an output-preference win (the retired §4 null).

In plain terms. §8 checked whether the critic’s *sources* are real and on point. The same tool, pointed at the *answer*, asks a simpler question of every number in it: does this follow from what we were given, or is it flagged as a guess? A number that is neither — stated as fact but resting on nothing — is the quiet kind of error that ships. A calculator, not a model, does the checking; and where a number was both wrong and load-bearing, fixing it flipped the decision to the one the numbers actually support.

9. What this establishes — governance by verifiable challenge

Put the three pieces together and the thesis is specific. SPARRING’s contribution, on the evidence here, is **governance of AI-in-the-loop decisions by verifiable challenge** — by *governance* we mean three checkable properties, not a softer “well-managed”: reasoning that is **consequential** (it catches real, would-have-shipped defects — §6), **grounded** (its critique stands on checkable artifacts, not confident assertion — backing $\approx 96\%$ of its concerns, far above a raw pass and modestly above even a competent skeptic, at no precision cost — §7), and **auditable** (whether a cited artifact *exists* is confirmable against public registries (MITRE, Crossref, the IETF datatracker) with no model in the loop; whether it is cited *correctly* is checked by a dual-vendor judge — across all

four defect families, and (exploratory) extended to the answer’s own load-bearing figures — §8). Transparency you can *trust*, because the grounding is **machine-checked for existence and cross-vendor-judged for application, not a single model grading itself**: a reader can confirm a cited CWE or DOI exists with no model and no domain expert. What stays model-coded is *legitimacy* — whether a flagged concern is genuinely real — which the verifier does not touch; it routes to the named domain-expert audit. This is not the claim that a blind reader prefers the *content* — that is the V2 null (§4), and it stands. It is a different and, we argue, more useful claim: an answer you can **act on** — its blind spots caught by a challenge you can verify, its unearned certainties gated — *better* in the dimension a decision-maker actually needs, even where it is not the one a blind taste-test measures. This is not a claim that the challenged answer carries fewer flaws than any alternative: on *net* would-ship flaws an effort-matched self-revision matched or beat it (§6.4) — but that self-revision hands back only a cleaner draft, with no way to tell whether it caught the defect that mattered. What earns the challenge its place is that what it surfaces is *verifiable*, not that the delivered answer is cleaner. That is exactly the case the method was built for, where an agreeable AI is in the decision loop and the risk is confident, well-formatted wrongness sailing through unchallenged. We stake that claim on the parts a reader can check without us — the real CWEs, the verifier’s mechanical resolution, the specific would-have-shipped catches — and offer the model-coded rates (§6.2’s base rate, §7’s legitimacy coding) as directional corroboration, not proof.

Four lines hold this together, and the paper fails if any of them blurs:

1. **Two axes of “better” — name which one, always.** An answer can be better in two senses, and this paper earns exactly one. On **blinded content-preference** (the output-preference axis of §3–§5, named precisely — with form flattened, what is left to prefer is the content) — a stranger shown two flattened answers preferring one — SPARRING is *null* (§4), and that stands as the honest report. On **trustworthiness and auditability** — blind spots caught by a *verifiable* challenge, unearned certainties gated, grounding a reader can check — it is genuinely better: a better answer to **act on**. The wall only needs to hold one way: the trust-axis gain must never be read as a blinded-preference win, because the moment it is, the retired output-preference claim sneaks back in and the V2 null reopens against it. So we name the axis every time — better to act on, not preferred on sight — and keep the two bodies of evidence in separate sections on purpose.
2. **“Verifiable,” not “obvious.”** A transcript is near-obvious; the contribution is not “you can see the reasoning” but “you can tell whether the reasoning is real.” The weight rides on the *verifiable* parts — the real CWEs a reader can check against MITRE, the verifier’s machine-checked grounding, the specific would-have-shipped catches confirmable without any LLM. The model-coded rates — §6.2’s base rate and the *legitimacy* coding behind §7’s grounding rates — are *supporting and caveated*, never load-bearing on their own.
3. **Auditable must be shown consequential.** Auditable theater is still theater. We answer that by leading §6 with the concrete catches — fail-open auth, the lockout DoS — and keeping them honest about survivorship. The governance claim earns its keep only because the audited reasoning *changed real decisions*.
4. **The claim is scoped to SPARRING’s Resolver.** The result in hand (§6.3) isolates a *single, ungated* challenge — so the established finding is that the adversarial challenge, at its simplest, is load-bearing. This report’s full subject is the Resolver at *both* depths: that single pass, and the both-must-agree gate that drives the challenge toward a revised, agreed output (or reports the disagreement) — the latter now measured in §6.4, a controlled test finding the gate adds no output-quality gain over a single challenge (a pre-committed null that reinforces

this paper’s thesis, not a content-quality gain). The **Proposer** — SPARK → Pattern Lock → the Cut — is a *separate* forthcoming report, not a gap in this one. So “SPARRING governs AI-in-the-loop decisions” means “its adversarial challenge does — consequentially and auditably, though iterating it to agreement buys no cleaner answer (§6.4),” and the Proposer’s value is left to its own report.

And the V2 null is not buried under all this. It is the **hinge**: the honest, pre-registered turn that sent the inquiry from the wrong question to the right one. A governance paper that *began* by claiming governance would be a reframe; this one earned the reframe by first running the obvious test, reporting that it found nothing it could resolve, and reading that result honestly enough to follow where it pointed — and where it pointed is visible in the frozen data (§5.1), not only in the method’s biography.

10. What the evidence implies for practice — the accountability pass

The established findings point to a specific, minimal way to run the Resolver on a real decision. We name it the **accountability pass** and offer it as a *design implication drawn from these results* — not as a locked empirical claim. The tags below carry the honest split: **established** = pre-registered and directly measured in this report (though still directional, §11); **exploratory** = not pre-registered (the relevant arms of the resolver-iteration pilot, and §8.1). Three of the four moves are exploratory, flagged below; the confirmatory test for them is named in §12 (item 8).

The name states the shape. The **pass** is a single grounded challenge, not a debate iterated to agreement. The **accountability** is what that pass is delivered with: a *definitiveness* gate (§8.1) against unearned certainty, and an honest map of how far to trust the result. Four moves:

1. **One grounded challenge, then stop.** A single adversarial challenge — ideally cross-vendor — recovers most of the producer’s own would-ship blind spots (§6.3); iterating it to agreement adds no output-quality gain and can churn in new flaws (§6.4). Run one well-grounded pass; do not iterate for quality. The pass earns its place because what it surfaces is *verifiable* — not because its answer is the cleanest available; on *net* would-ship flaws a quiet effort-matched self-revision matched or beat it (§6.4), but self-revision leaves no checkable trace of what it caught, so you cannot tell whether it caught the defect that mattered — and that trace is the point. (*Established*: §6.3, §6.4.)
2. **Gate every hard specific for definitiveness.** Each number, date, or named quantity in the answer must either trace to the problem’s given facts or be marked as an estimate or assumption; an unmarked, underived specific fails the answer *even if it happens to be true*, because the system cannot know which fact the user’s decision will turn on (§8.1). (*Exploratory*: §8.1.)
3. **Return two parts: the answer, and a trust rider.** Keep the recommendation clean; put the accounting — what is grounded, what is assumed and should be checked against the user’s own situation, what was contested and held, and an overall confidence — in a short, separate rider. Forcing that accounting into the answer itself *worsened* the recommendation on our set — the interleaved arithmetic and hedges appear to blunt the recommendation’s clarity — so splitting the channels keeps the answer sharp and gives the extra scrutiny a job. (*Exploratory*: *resolver-iteration pilot; results below*.)
4. **Fix by anchored correction.** When the gate flags a figure, revise it in place — correct that specific figure against its supporting evidence, leaving the rest of the recommendation standing — rather than re-deriving the whole decision from scratch: anchored correction was judged better-or-equal on all four flagged cases (n=4) by both vendors and reduced unsupported

figures, where re-deciding fresh introduced new ones. (*Exploratory: resolver-iteration pilot; results below.*)

The two delivery elements without a home elsewhere in the body rest on the resolver-iteration pilot (n=12 realistic decision problems, no answer key; scored by cross-vendor agreement plus calculator checks, so read them as directional). **Two-part output:** when the accounting was forced *inline* — the author made to show all its arithmetic while drafting — unmarked shaky figures fell, but the recommendation was judged better in only 1 of 12, so the accounting moved to a separate rider (move 3). **Anchored correction:** revising the flagged figures in place was judged better-or-equal on all four flagged cases by both vendors and cut unsupported figures 8→5, where re-deciding fresh from the findings introduced them (8→10) (move 4). Both results are exploratory and directional; their pre-registered confirmation is §12 (item 8).

This is the operating form of *governance by verifiable challenge*: one grounded challenge, a gate that refuses unearned certainty, and an answer handed over with a checkable account of how far to rely on it — the trust axis of §9's distinction, made operational: an answer whose grounds a decision-maker can inspect before acting.

11. Validation stance and limitations

11.1 Validation stance — an LLM-validated first signal, with a named path to expert-grade

We state plainly what kind of evidence this is. The §6 results (the catch base-rate and the controlled catch-rate) are an **LLM-validated “first signal”** — coded by knowledgeable models, adversarially re-judged to correct leniency — resting for their *verifiable* core on the real would-have-shipped catches a reader can confirm directly. §7's **grounding** rates are stronger: machine-checked by the §8 verifier (existence model-free, correct-application cross-vendor), no longer one model coding another. What is **not** yet human-adjudicated, on either, is **legitimacy** — whether a flagged concern is genuinely real. We chose this stance deliberately. The seeded-defect cases reach into technical domains — specific CWEs, I-O-psychology citations — that the author cannot personally adjudicate — there is no human answer key, the **no-gold constraint** this paper works under — so *author-as-domain-auditor was the wrong design*. Instead: the **verifier** (§8) makes the *grounding* axis objective without domain expertise; *legitimacy* on technical specifics routes to a **future domain-expert audit**, not to the author; and the author's role is to **decide the validation strategy and vouch for the honesty of the framing**, not to pass a technical exam. The expert-recruitment path is named future work, not a present claim.

11.2 Limitations

1. **The output-preference study is underpowered by design** (two cases, ten raters) and has **near-floor inter-rater agreement** (§4.4) — a calibration check, not an effect-size estimate; it cannot conclude SPARRING is worse, only that it could not resolve a preference difference. *Attaches to: the pivot (whole Resolver) — Surfaced in: J1 · Output-preference study — a blinded taste-test could not resolve a difference, so it redirected the inquiry rather than measuring SPARRING worse.*
2. **One completer's prior-familiarity covariate is missing, not imputed** (§3.4). Of the ten who finished both cases, one never submitted the post-submission familiarity form; the gap is disclosed rather than silently defaulted, and that completer is excluded — not assumed F0 — from the low-exposure (F0–F2) sensitivity subset, which stands at n=9 rather than n=10.

The gate’s FAIL holds under every familiarity-based cut, including this one (§4.6). *Attaches to: the pivot (whole Resolver) — Surfaced in: J1 · Output-preference study — a missing covariate on one completer is disclosed and excluded, not imputed; the gate’s result is unchanged under every familiarity cut.*

3. **The mechanism audit is survivorship-biased (§6.2)** — its 67% is conditional on a ceremony being substantial enough to record, and that is the dominant caveat; it is also small-n and LLM-coded. *Attaches to: S1 · one grounded pass — Surfaced in: J3 · Mechanism audit — the 67% is conditional on recorded ceremonies; read it as directional, not a population base rate.*
4. **This report covers the Resolver, not the Proposer.** SPARRING has two parts: a *Proposer* (the SPARK → Pattern Lock → the Cut front end, which produces the proposal) and a *Resolver* (the adversarial challenge that works on it). This is the Resolver’s report. Whether the Proposer earns its cost over a naive proposal is a *separate forthcoming report* (§12, item 5), not a deficiency in this one — so this report does not measure the marginal value of SPARK, Pattern Lock, or the Cut. The one phase-level signal that surfaced here (§6.2’s adversarial re-judge downgrading SPARK-phase wording polish as non-load-bearing) is noted, not load-bearing on this report’s claims. *Attaches to: report scope (the Resolver, not the Proposer) — Surfaced in: J2 · the pivot (which scoped the report to the Resolver) — the Proposer’s value is a separate report (§12), not a gap here.*
5. **The gate’s controlled result is a pre-committed null (§6.4).** SPARRING’s both-must-agree, non-bypassable gate (§2) would drive a challenge **toward** a revised, agreed output (or report the disagreement). Isolated on a fresh dependent variable (net material flaws remaining, new flaws counted), iterating to agreement produced no cleaner output than a single challenge and did not beat an effort-matched self-revision control — a null on output quality that reinforces this paper’s thesis. It is directional (n=8; the pre-registered expansion trigger to ~16 problems is met and carried as future work, §12), and it establishes the gate’s value as governance and the non-bypassable *design*, not a quality gain; at n=8 the absence of a fake-agreement signal is absence of evidence, not proof that the gate resists being gamed into agreement. *Attaches to: P3 · the retired both-must-agree gate → S4 · trust rider — Surfaced in: J5 · Resolver-iteration — iterating to agreement is an output-quality null; the gate’s worth is governance and the non-bypassable design, not a cleaner answer.*
6. **The gate-discrimination result (§6.5) is exploratory and a forming negative result, not yet clean.** Across five gate variants an LLM admissions gate’s block *rate* tracks wording while its *discrimination* of material regressions stays at chance (balanced acc ≈ 0.50 – 0.59 , $\kappa \approx 0.00$ – 0.19), and a frontier-model re-run (claude-fable-5, two ways, zero refusal/fallback contamination) rules out model capability as the cause. But one confound remains open: the net-flaw labels are ± 1 -noisy and only 11 of 61 rows are regressions, so a flat result cannot yet separate “no model can discriminate” from “the labels are noise.” The claim is therefore bounded to *framing controls rate, not discrimination, and the wall is not a capability ceiling* — the label-noise cross (§12) is required before “LLM gating cannot reliably discriminate regressions” is asserted. It is exploratory, not pre-registered, on a small synthetic set. *Attaches to: S1 · one grounded pass (the gate program tested whether to bolt automated admission-gating onto it) — Surfaced in: J5 · Resolver-iteration — an admissions gate’s block rate is a wording artifact and its discrimination is not fixed by the frontier model; the label-noise confound keeps the negative result forming, not final.*
7. **The definitiveness gate (§8.1) is exploratory and not pre-registered.** It extends the grounding check to the answer’s own figures; its traceability verdict is mechanical (a calculator on quoted inputs), but the *presentation* call is a cross-vendor model judgment (~90% agreement — a shared rubric, not correctness) with splits escalated, and precision (~9 of 11 descriptive

- flags genuine) is directional at $n=12$. Its decision-flip demonstration is a **constructed positive control** (three single-threshold cases, calculator as judge), not a base rate. *Attaches to: S2 · definitiveness gate — Surfaced in: J5 · Resolver-iteration — the traceability verdict is mechanical, but the presentation call is a cross-vendor judgment, directional at $n=12$.*
8. **The seeded-defect grounding is machine-checked (Path-A) but the pilot is small and synthetic** (§7, §8): on the v3 corpus the verifier resolves grounding mechanically across all four families (existence model-free, correct-application cross-vendor), so the rates no longer rest on one model coding another. The live caveats are $n = 99$ concerns / 12 cases (directional, no inferential significance), LLM-generated cases, same-vendor *ceremony* agents, and $\sim 2\times$ cost for the two-agent condition. What is **not** machine-resolved is **legitimacy** — whether each flagged concern is genuinely real — which routes to the domain-expert audit (§11.1), not the verifier. *Attaches to: S1 · one grounded pass — Surfaced in: J6 · Seeded-defect grounding pilot — grounding is machine-resolved, but legitimacy is not, and the corpus is small and synthetic.*
 9. **The verifier’s two known tool-side gaps** — surfaced on the earlier external-canon run (§8) — run in opposite directions: one over-credited a rhetorical CWE name-drop as grounding, one failed to resolve an under-specified author-year citation (it needs the title for Crossref); neither is a fabrication, and neither overturned the coding’s direction. *Attaches to: S1’s grounding verifier — Surfaced in: J7 · the grounding verifier — the two failures are small, two-sided, and non-fabricating, so they don’t bias the grounding rate in either direction.*
 10. **The v2 corpus’s machine-checking ceiling is closed in v3.** The earlier (v2) seeded-defect corpus could machine-check only its $\sim 7\%$ external-canon slice, because its per-case briefings were never persisted (only answer-keys were saved), so the in-briefing majority stayed expert-coded — a data-loss artifact, not a design limit. v3 closes it by persisting every briefing, so grounding is now machine-resolved across families (§8). The residual ceiling is therefore no longer *coverage* but *legitimacy* (above) and *scale* (small- n , synthetic cases). (Note: the verifier-ceiling figure depicting the v2 $\sim 7\%/19\%/57\%/17\%$ breakdown is superseded by this and should be re-cut from v3 before any use.) *Attaches to: S1’s grounding verifier — Surfaced in: J7 · the grounding verifier — v3 closed the coverage ceiling; the residual limit is legitimacy and scale, not coverage.*
 11. **Domain reach of the model-free check.** The fully model-free, no-expert existence check is strongest where a domain has a public registry to resolve against (security→MITRE/CWE, scholarship→Crossref/DOI, standards→IETF); each such registry is a **wired backend added once by the operator, not discovered at run time** (§8). The dominant in-briefing check (86 of 99, §8) is domain-independent — it uses the briefing the decision already supplies — and the seeded families span both in-briefing and external checks, so the *mechanism* is not security-specific. But in a domain with no external canon, out-of-document claims lose the model-free guarantee and fall back to the expert legitimacy audit, and §7.3’s largest challenger edge sat on the external families — so the most dramatic, no-expert gains are registry-dependent even though the method is not. One precision asymmetry follows from the same design: an unresolved literature search returns *refuted*, not *unchecked*, so a real but unindexed source — a self-published manual, a pre-modern primary source — can be **false-refuted**; registry-absence is not proof of fabrication, and a defer-when-no-registry-matches path is the named fix. These pilots exercise only the software/security, scholarship, and standards registries; a non-registry domain is untested here. *Attaches to: S1’s grounding verifier — Surfaced in: J7 · the grounding verifier — the model-free existence guarantee is registry-dependent; non-registry domains fall back to the expert audit, and an unresolved search false-refutes rather than defers.*
 12. **Single substrate in these pilots — a measurement gap, not a method gap.** The study, the mechanism audit, and the seeded-defect pilot all ran on one vendor (Claude); the verifier’s

supports-claim judge and the study’s judge panel were cross-vendor, but the *ceremony agents* were not. SPARRING’s design prescribes — and the author uses, for high-consequence decisions — **cross-model** agents: Generator and Challenger on different vendors. This pushes the method’s disjoint-evidence discipline (the two roles work from non-overlapping evidence) one level deeper — from the *evidence* the agents read to the *models* running them. Same-vendor pairings share a training distribution, RLHF (reinforcement learning from human feedback), and sycophancy-reduction tuning, so the two can drift into *coordinated* theater — agreeing for shared trained-in reasons rather than on the merits. Running the roles on different vendors is designed to **reduce (not eliminate)** that residual risk. Whether cross-model *measurably* reduces that risk is structurally argued, not isolated in these single-substrate pilots; the controlled test is named future work (§12). *Attaches to: all ceremony agents (whole Resolver) — Surfaced in: J3 / J4 / J6 (all ran single-vendor) — cross-model agents are prescribed but untested here; the controlled test is future work (§12).*

13. **Recruitment channel** for the rater study was a convenience sample from the author’s social media — selection bias disclosed, not assumed away. *Attaches to: J1’s rater pool — Surfaced in: J1 · Output-preference study — a convenience sample; selection bias is disclosed, not assumed away.*

12. The path forward

The redirect names its own confirmatory work. In priority order:

1. **Scale and harden the verifier-backed grounding (Path-A).** The v3 run already lifted grounding onto Path-A **across all four families** (§8) — existence model-free, correct-application cross-vendor, in-briefing included — by persisting every per-case briefing, closing the data-loss gap that had held v2 to its ~7% external slice. What remains is **scale** (beyond $n = 99$ concerns / 12 cases), **hardening** the two extraction gaps the run surfaced (rhetorical-mention vs. grounding-citation; title-aware citation resolution), running the claim→fact mapper **dual-substrate** for high-stakes use, and **atomizing concerns** so in-briefing support-judging stays trustworthy at scale. Persisting per-case briefings is now standard practice, not a fix to schedule.
2. **A prospective, survivorship-free mechanism ledger (built, live).** Every SPARRING ceremony now self-codes whether the Challenger was load-bearing and appends to a per-project ledger plus a global roll-up. Within weeks this yields the **unbiased all-ceremonies base rate** — trivial spars included — that §6.2’s survivorship-limited audit can only approximate. Because runs self-score optimistically, the honest number requires periodically re-running the adversarial refute-by-default re-judge over the accumulated log; the tooling says so in its own output.
3. **A domain-expert audit** of the seeded-defect coding’s *legitimacy* axis, per the validation stance (§11.1) — the expert-grade complement to the verifier’s objective grounding axis.
4. **A pre-registered process study.** Measure what the output-preference study could not: does running SPARRING change the practitioner’s decision, catch errors they would otherwise have missed, or move their calibrated confidence — a within-decision, process-level design.
5. **The Proposer report (a separate forthcoming paper) — does refining the input earn its cost?** This report settles the Resolver; the *Proposer* — SPARK + Pattern Lock + the Cut — gets its own. Those three phases act as one unit, not separable steps: SPARK (expansion) is only useful once Pattern Lock filters it, and a filtered set is only useful once the Cut decides it. The question: does a Proposer-refined proposal beat a naive one once handed to the (now-characterized) Resolver? The design **builds up from the validated core rather than**

dismantling the whole, run as a factorial (Proposer on/off × Resolver setting) so it yields the Proposer’s value, the full SPARRING flow, *and* two results carried as first-class, not robustness afterthoughts: the **interaction** — whether a refined input makes the challenge add more or less (a cleaner proposal may leave the challenge less to catch, making the Proposer partly *redundant* with the Resolver) — and a **compute-matched control**, a naive proposal given equivalent extra passes without the Proposer’s structure, so any gain is attributable to structure rather than effort (the §7.2 lesson, one level up). It either justifies the full method or holds the established claim scoped to the Resolver (§9). The §6.3 harness supplies the resolver and audit-panel machinery; the new work is generating and scoring the refined-input conditions.

6. **Make the verifier a standing ceremony step** — turn the pilot-local tool into a reusable component run *within* the ceremony, so grounding is checked at ceremony time, not only in post-hoc pilots; this includes the answer’s-own-figures definitiveness gate of §8.1, whose exploratory status a standing, at-ceremony tool would help move toward confirmatory.
7. **Test and tool cross-model SPARRING**. Run the same ceremonies with Generator and Challenger on **different vendors** against the same corpus, to isolate whether cross-model agents measurably reduce residual coordinated-theater beyond the same-vendor, disjoint-evidence default (the specification’s cross-model integration stage). Add **per-agent vendor selection** to the ceremony runner so the cross-model variant the method already prescribes — and the author already runs manually — is automatable, not only hand-orchestrated.
8. **Confirm the accountability pass as an operating profile (§10)**. The pass’s load-bearing move — a single grounded challenge — is established (§6.3, §6.4); its three delivery elements are exploratory and need a pre-registered confirmatory run. (a) **The definitiveness gate’s instrument**: how reliably an automated bucket-classifier (traceable / flagged / neither) agrees with careful human bucket-labels across a larger realistic corpus — its precision on descriptive flags is directional at n=12 (§8.1). (Those are labels for *classifier* QA, not a gold *decision* answer.) (b) **The two-part output**: whether splitting the recommendation from a separate trust rider preserves answer quality — scored on the model-established net-flaw instrument (§6.3/§6.4’s audit panel, inheriting its directional caveat), **not** a preference judgment, since §4 showed cross-vendor judges do not track human answer-preference — while improving calibrated accountability, tested against the single-part control this pilot found *worse* when the accounting was forced inline (judged better in only 1 of 12). (c) **Anchored correction vs. deciding fresh**: the gate-revise contrast (anchored correction better-or-equal on 4 of 4 flagged cases, unsupported figures 8→5, against 8→10 for re-deciding from the findings) re-run at scale with replicate scoring. All three are decidable without a gold *decision* answer — the gate on each hard specific’s own bucket, the net-flaw contrasts on the model-established audit panel plus calculator checks — so the no-gold constraint (§11.1) does not block them; the *answer-quality* half of (b) still inherits §4’s judge-alignment caveat and is read as directional. With item 6 (the verifier as a standing ceremony step), this is what would move the §10 profile from design implication to confirmed practice.
9. **Independent replication**. Every result in this report was produced, audited, or adjudicated by the author’s own tooling, on the author’s own codebases, or by raters the author personally recruited (see the Conflict of Interest statement, following §14). None of it has yet been replicated by a party with no relationship to the author or Resource Forge LLC. That replication — ideally on decisions and codebases the author did not produce — is the one check this report cannot perform on itself, and it is named here rather than left implicit.
10. **Close the gate-discrimination negative result (§6.5)**. The exploratory finding — an LLM admissions gate’s block *rate* is a wording artifact while its *discrimination* of material regres-

sions is not, and the frontier model does not repair it — has one open confound: the net-flow labels are ± 1 -noisy, so a flat result cannot yet separate “no model can discriminate” from “the labels are noise.” The decisive step is to **replicate-score the 61 champion→candidate pairs** ($k \approx 5$, **averaged**) and re-run both model tiers on the de-noised labels; only if both stay at chance on clean labels is the “LLM gating cannot reliably discriminate material regressions” claim publishable. If the de-noised regressions are real, the one untested lever — **labeled few-shots mined from gate-vs-truth disagreements** — is tried before the negative result is written. Either outcome reverts the program to the validated single-challenge kernel (§6.3); this item decides whether §6.5 firms from *forming* to *established*.

An open invitation. This report is a first signal from a small software studio, not a lab’s final word — deliberately so. SPARRING was built to solve a live problem in the author’s own software, and the empirical work here is a modest extension of that production need, run at the scale a micro-business software studio can support — not the powered, multi-site study the strongest of these claims ultimately warrant. That larger study is precisely what a better-resourced group — an academic lab, an evaluation team, a well-tooled research org — is positioned to run and a micro-studio is not. The groundwork is already public and reusable: the pre-registration is locked (§13), the corpora, analysis code, and grounding verifier with its self-tests are archived and browsable, and the confirmatory designs above are specified in enough detail to lift directly. We would welcome that work — including replication mounted specifically to *refute* these findings — and will support serious attempts at it. The most useful thing this paper can do is not to settle the question, but to make it cheap for someone better-equipped to settle it.

The lesson is not “stop testing SPARRING.” It is “test the claim the method actually makes” — and the claim, the data now say, is about governance.

13. Data and code availability

All public materials are archived at **Zenodo** (concept DOI [10.5281/zenodo.21210265](https://doi.org/10.5281/zenodo.21210265), resolving to the latest version) and browsable at github.com/muddyone/sparring-publicaccess, under papers/2026-resolver-governance/: the pre-registration and its amendment ledger, plus the original locked text (study-output-preference/pre-registration-LOCKED-2026-05-03.md); the frozen aggregate results and analysis code (study-output-preference/analysis/.../scripts/); the mechanism-audit corpus and both coding passes (governance/mechanism-audit/); the seeded-defect grounding pilot with the grounding verifier and its self-tests (governance/seeded-defect/); and the resolver-iteration pilot behind §6.4 (governance/resolver-iteration/). The SPARRING framework specification and guides are under framework/.

Exposure posture. Raters are de-identified (labels E1–E5 and user ids; no names or emails). The mechanism-audit corpus is the author’s own proprietary codebases; only *derived* findings and codings are published, never source — raw material is available from the author on request. Bulky, unaudited per-run transcripts for §6.4 are omitted; the codings, results, and code reproduce every reported number. The blind-rater instrument runs at resourceforge.com/sparring; its source is not public.

14. Acknowledgments

We thank the ten raters who each gave an hour to an AI-evaluation study in which they had no personal stake in the outcome. Per each rater’s recorded consent preference, we acknowledge by

name Rene Cope, Matthew Niedner, Kelly Cope, Vivian Farris, Sally, Michelle, Missy Borchardt, and Danomorr; two further raters preferred to remain anonymous.

Conflict of interest

The author is the founder of Resource Forge LLC, with which SPARRING is publicly associated (§1.2; also disclosed in the author’s own social-media posts about the method) — a financial and reputational interest in the method’s apparent value, not an incidental affiliation. The output-preference study’s raters (§3.4) are the author’s own personal and professional contacts, reached through his own social media, not an independent panel; the mechanism-audit corpus (§6) is drawn from the author’s own commercial codebases. Pre-registration (§3.1) and the pre-set decision gate are the controls built to bound this conflict for the one test run on people with no stake in the outcome — and that gate failed (§4), which is at minimum evidence it was not gamed after the fact. The mechanism-audit and seeded-defect evidence, evaluated on the author’s own material, are not protected by that same control and are read most safely as motivated first-signal pending the independent replication named in §12.

Generative-AI use and author responsibility

This manuscript was prepared with substantial assistance from generative AI, disclosed here in full — consistent with the paper’s own argument that AI-assisted reasoning should be auditable rather than hidden.

Tools. Manuscript preparation and the SPARRING review agents used Anthropic’s Claude (Opus 4.7/4.8) via Claude Code. Verification used a second independent vendor by design: the grounding verifier’s supports-claim judge and the cross-substrate citation-verification engine run on **Claude and OpenAI**, and the V2 study’s judge panel used three vendors (**Claude, OpenAI, and xAI’s Grok**; §3.3).

Manuscript preparation. AI assisted with drafting and editing the text, with building the analysis and verifier code (§8, §13), and with structured review passes — argument-and-evidence, accessibility, and prose craft — that surfaced specific revisions the author then adjudicated. One such pass, an adversarial two-agent SPARRING ceremony on §8’s central claim, surfaced the existence-vs-correct-application distinction the author then adopted. This is reported as disclosure and as an honest illustration of the workflow; it is **not** offered as evidence that the method works (that would be circular), and the paper’s empirical claims rest solely on §4–§8.

Evidence and methodology. AI’s role in the *research* — distinct from manuscript preparation — is part of the experimental design and is disclosed in-line throughout: the decision materials and both answers were generated by Claude Opus 4.8 under both conditions (§3.2); the judge panel is a cross-vendor AI ensemble (§3.3); the mechanism-audit rates are LLM-coded (directional; §6), the seeded-defect *grounding* is machine-checked (Path-A; §8) with legitimacy expert-pending, and the grounding verifier uses a dual-substrate (Claude + OpenAI) judge (§8). The limits of the model-coded evidence are stated where that evidence is used (§7.4, §11).

Responsibility. The author directed the work, reviewed all content, and verified what is mechanically checkable; the domain-technical legitimacy he cannot personally adjudicate is routed to the named future expert audit, per the validation stance (§11.1). The author takes full responsibility for every claim, number, and citation in this paper. The AI tools are not authors and bear

no accountability; per prevailing publication norms, generative-AI systems cannot be credited as authors.

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Appendix A — pre-registration deviation ledger (output-preference study)

Date	Deviation	Pre-data?
2026-05-03	Condition-B rendered format-symmetric to Condition A (blinding)	yes
2026-05-27	Rater pool opened to public convenience sample	yes

Date	Deviation	Pre-data?
2026-06-04	Plain-language rebuild; substrate 4.7→4.8 both	yes
2026-06-04	conditions; judges re-run Naming/identity blind; familiarity covariate; partial-completer handling; audio covariate	yes
2026-06-12	C5a re-scoped to closed-world factual accuracy; judges re-run	yes
2026-06-14	Cross-case load covariate; case-order counterbalancing; deferred simultaneous un-blinding	yes
2026-06-24	n-keyed recruitment extension (early close at $n \geq 10$ both cases; hard 2026-07-01); closed 2026-06-26	decided blind
this report	Primary judge-alignment reference = unaided completers (n=6), since the pre-reg “unaided RF-partner” reference collapsed to n=1	post-data, disclosed

Appendix B — the evidence map (what each piece proves, and what it does not)

Evidence	Proves	Does NOT prove
V2 rater null (§4)	output-preference can't see process value; the honest pivot	that SPARRING is worse (uninformative, not negative)
Mechanism audit (§6)	the process is consequential — catches externally-checkable defects that would have shipped	a population base rate (survivorship-limited)
Seeded-defect pilot (§7)	the critique is grounded in checkable artifacts (machine-checked, directional, small-n), non-theater, at no precision cost; far above a raw pass, modestly above a competent skeptic	better <i>outcomes/answers</i> (retired); the <i>legitimacy</i> of each concern (expert-pending)

Evidence	Proves	Does NOT prove
The verifier (§8)	a cited artifact’s existence is machine-checkable (no model, no domain expert) across all four families on the v3 corpus (86/6/7 of 99); correct-application judged cross-vendor	that a flagged concern is legitimate (routes to the domain-expert audit); generality beyond this small, synthetic corpus
Gate-discrimination probe (§6.5)	framing controls an LLM admissions gate’s block <i>rate</i> but not its <i>discrimination</i> of material regressions (balanced acc $\approx$ 0.50–0.59, $\kappa \approx$ 0.00–0.19 across five variants); the wall is not a model-capability ceiling (frontier-model re-run flat two ways, zero contamination)	that LLM gating <i>cannot</i> discriminate regressions (label-noise confound still open); exploratory, not pre-registered; small-n synthetic
Iterate-to-agreement study (§6.4)	the both-must-agree gate adds no net-flaw quality over a single challenge (pre-committed null; self-revision matches it) — reinforces the governance-not- <i>blind-preference</i> thesis	that the gate is <i>gamed</i> (no fake-agreement signal, but absence of evidence, difficulty-confounded); a base rate (n=8, directional)
Definitiveness gate (§8.1, exploratory)	mechanical grounding generalizes from cited artifacts to the answer’s own stated figures; (constructed control) a decision-pivotal wrong figure is caught and correctable	a base rate, better answers, or that the presentation call is gold (cross-vendor ~90%, calibration-set, splits escalated)

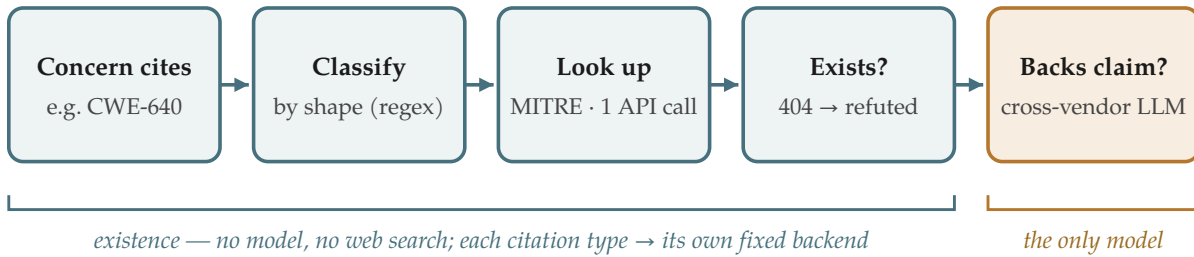
Appendix C — the verifier, end to end

§8 reports *what* the grounding verifier checks and how far it reaches; this appendix shows *how it decides*, so the “machine-checked, no model in the loop” claim can be inspected rather than taken on trust. The pipeline is deliberately simple, and that simplicity is the point: the existence check is a classifier plus a lookup — no language model and no open-web search.

The steps, in order:

1. **Classify by shape (regex, no model).** The classifier keys off the *form* of the citation string — CWE-\d+, RFC \d+, a DOI / arXiv id or an Author YYYY, an https://... URL, a file.ext:line, an in-briefing #3 / §2 — not the subject matter. It routes to exactly one backend, or to *ungrounded* if nothing matches.
2. **Resolve existence (one hardcoded call — no model, no web search).** MITRE’s CWE API and the IETF datatracker are direct id lookups (a 404 refutes a fabricated id); Crossref is a bib-

How one concern is checked, end to end



The classifier reads the citation's *type*, never the topic; a citation matching no backend is counted ungrounded.

Figure 7: One concern through the grounding verifier. A regex classifier reads the *shape* of the artifact the Challenger cited — not its topic — and routes it to a fixed backend; each backend does a single, model-free resolution: MITRE's API for a CWE id, the IETF datatracker for an RFC, a Crossref search for an author-year/DOI, a working-tree read for a `path:line`, a live fetch for a URL, a text-match against the supplied briefing for an in-briefing `#fact/$section`. Existence is settled with no model and no web search; only then does a single model step run — the cross-vendor (Claude + OpenAI) judge that rules whether the resolved source actually backs the claim. A citation matching no backend is counted ungrounded, never passed silently.

liographic *search* matched on author surname + topic + year; a code reference is read/greped against the working tree; a URL is fetched read-only (a dead link refutes); an in-briefing reference is matched against the persisted briefing text. An unresolvable artifact is a *durable* refuted/unchecked state, never silently upgraded.

3. **Judge application (the only model — cross-vendor).** For an artifact that exists, a second check asks whether it actually *backs* the claim it was cited for. Per the project's cite-check rule this runs on two independent vendors (Claude + OpenAI) and surfaces disagreement rather than forcing convergence.
4. **Legitimacy is not attempted here.** Whether a flagged concern is genuinely *real* is a third axis the verifier does not touch; it routes to the domain-expert audit (§11.1; §12, item 3).

Two consequences worth stating plainly. Because routing is by citation *shape*, the check is **domain-general in mechanism** but **registry-bound in its model-free reach**: a domain with a public registry (security→MITRE, scholarship→Crossref, standards→IETF) gets a fully model-free existence guarantee, while a domain without one leans on the in-briefing, code, and URL backends and routes the rest to the expert (§11.2). And because an unresolved literature search returns *refuted* rather than *unchecked*, a real-but-unindexed source can be **false-refuted** — a known precision gap with a named fix (defer when no registry matches; §11.2). The engine is the research-origin `verify_grounding.py` in the seeded-defect pilot's `scripts/`; its self-tests cover every source type and the live-URL fetch.

Appendix D — Anticipated objections, answered plainly

This appendix is an invitation, not a dismissal. Each objection below is stated as sharply as we could make it — not a strawman we can easily beat — because a softened version of a critique defeats the purpose of naming it. Where the honest answer is a concession, we say so.

One more piece of context before the objections themselves: this paper’s vector into the literature is not the usual one. The author is not an academic; he is a practitioner who built SPARRING to solve a real operational problem in his own commercial software, found it worked well enough in use to suspect it might carry a broader lesson, and wrote this report to test that suspicion rather than to close a gap he had found in someone else’s literature review. That origin shapes both what this paper is good at (concrete, checkable, real-world catches; a method under actual production pressure before it was ever measured) and where it is weakest (no institutional review, no co-author to disagree during drafting, and — see the Conflict of Interest statement, following §14 — an author with a direct stake in the answer). We would rather name that vector plainly than let a reader discover it and wonder why we did not.

“n=8, n=10, n=12 — this is far too small to support any of these claims.” Largely agreed, and said throughout the body, not just here: every result built on these samples is tagged directional, and none is offered as an effect-size estimate. The one place we would push back: several of these findings do not rest on the *sample size* at all — the five would-have-shipped catches (§6.1) and the real CWE numbers a reader can check against MITRE directly are true or false regardless of n. The small-n statistics establish *rate* and *direction*; the verifiable catches establish that the mechanism does something, at n=1 each. Conflating “the rate is uncertain” with “nothing here is established” would miss that distinction — but the rate claims themselves, taken alone, deserve exactly the skepticism this objection extends.

“You built SPARRING, you run a company associated with it, and your raters are your own personal and professional contacts — how is this not hopelessly conflicted?” The financial and personal stake is real, disclosed, and does not go away because it is disclosed. What we can offer is not a denial of the conflict but a description of what was built to survive it despite the conflict: the rater study was pre-registered before any data existed, specifically so the author could not shape the analysis after seeing results; the gate that failed (§4) was locked in advance and failed anyway, which is at minimum evidence the author did not rig the one test he ran on people with no stake in the outcome. The mechanism-audit and seeded-defect evidence is a different story and this objection lands harder there — those corpora are the author’s own ceremonies and the author’s own commercial codebases, and no amount of adversarial re-judging changes whose work is being graded. That is why independent replication (§12, item 9) is not a formality; it is the actual answer to this objection, and it does not yet exist.

“Why trust a solo, non-peer-reviewed preprint over work from an academic lab?” You should not trust it *more* — arXiv preprint status carries none of peer review’s guarantees, and this one is additionally solo-authored outside any institution, which removes even the informal check of a co-author disagreeing during drafting. What we would ask a skeptical reader to separate is the *source* of a claim from its *checkability*: a claim like “this CWE id resolves against the MITRE registry” is true independent of who wrote the sentence, and the paper is built so that as much as possible sits in that category rather than the “trust the author’s judgment” category. Where a claim cannot be moved into the checkable category — the mechanism-audit rate, the seeded-defect legitimacy coding — we have said so plainly rather than dressed it up as more than it is. That is a mitigation, not an answer to the underlying point: solo, unreviewed work is exactly as reliable as its verifiable

parts, and no more.

“This is enormous and runs six overlapping numbering systems — why should a reader wade through all of it?” No good defense of the length exists beyond “the honest version of this story had this many moving parts, and cutting any one of them would have meant asserting instead of showing.” The numbering systems are a fair complaint on their own terms, independent of length — J#/S#/P#/V#/v#/R# is a lot to hold at once, and we have added a consolidated key (§2.3) to cut the cost of that choice without pretending the choice was free.

“The headline result is a null — why does that deserve a full paper rather than a short note?” Because the null is not the paper’s contribution; the redirect it forced is. A short note reporting “no preference found” would have been a smaller, truer-to-genre paper and a worse account of what actually happened: the null revealed that the instrument, not the method, was blind to what mattered (§5.1), and that diagnosis is only visible by walking through what the blind removed and what survived it. We accept that this makes for an unusual paper shape; we would rather ship the unusual shape than a conventional one that hides the reasoning behind the reframe.